

THE TOPOLOGY IS THE SINK: TOPOLOGICAL HAL-LUCINATION DETECTORS REDUCE TO FIRST-ORDER ATTENTION STATISTICS

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Paper under double-blind review

ABSTRACT

Persistent homology (PH) of attention graphs underlies a growing family of hallucination detectors, including the recent TOHA (ACL 2026). We show that, in decoder-only models, these topological features reduce to first-order attention statistics dominated by the *attention sink*, either exactly or up to one computable number. For any symmetric dissimilarity and vertex s , we define the *coning defect* δ_s . Combining classical cone and stability arguments, we show that the entire Vietoris–Rips barcode lies within bottleneck distance δ_s of a “star diagram” read off the distances to s . Every stable vectorization is therefore a function of one column of the attention matrix, up to δ_s . Applied to TOHA, the same argument shows that its prompt–response topological divergence *equals* the mean, over response tokens, of one minus their largest prompt attention whenever a prompt-level coning defect vanishes, and lies within that defect otherwise. On 18,906,944 per-head graphs from 16 settings (7 open models of 1.1B–7B parameters; TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark we build), this identity has no violations and is exact on 57–88% of them. Adding TOHA’s score to its first-order counterpart changes AUC by at most 0.007 (equivalent within ± 0.015 in 15 of 15 settings), and supervised probes on the two are equivalent in 15 of 15. For whole-graph PH, 838,356 layer graphs show no bound violations; in every model with a first-token sink, 36–98% are *exactly* coned, 1D persistence is zero on all of them, and length-normalized 0D persistence keeps a median per-layer rank correlation of at least 0.98 with sink attention. For detection, on TruthfulQA and on-policy TriviaQA, none of 0D PH, sink-deflated PH, or the per-head coning defect adds more than 0.009 AUC to cheap non-topological probes, and TOHA adds at most 0.013. We release the extractor and the evaluator, together with per-example features for a subset of settings.

1 INTRODUCTION

A growing family of hallucination detectors treats a transformer’s attention matrix as a weighted graph and summarizes it with persistent homology (PH): barcodes of connected components (H_0) and loops (H_1) along a Vietoris–Rips filtration (Kushnareva et al., 2021; Cherniavskii et al., 2022; Perez & Reinauer, 2022; Kostenok et al., 2023; Bazarova et al., 2026; Samaga et al., 2026). The appeal is that topology might capture *global* structure in how information flows that token-level statistics miss. A separate line of work has established that decoder-only language models route a large share of attention to the first token, the *attention sink* (Xiao et al., 2024; Sun et al., 2024; Cancedda, 2024; Gu et al., 2025; Barbero et al., 2025). Only recently have the two been connected: Binkowski et al. (2026) show that sinks carry much of the signal used by attention-based hallucination detectors. No work has yet asked what sinks do to the *topology* of attention graphs.

We show that, for topological detectors, the two literatures study the same object to a large and quantifiable extent. When one vertex s is closer to every other vertex than those vertices are to each other, every sublevel set of the Vietoris–Rips filtration is a cone with apex s : all higher homology vanishes and the minimum spanning tree (which computes H_0) is the star around s . Real attention graphs sit at or very near this configuration, because of the sink. The “topology” of a causal attention graph is therefore, in most layers, exactly a function of how much attention each token sends to the

sink, and in the remaining layers is controlled by a single computable number, the coning defect. The same holds for the published detector that restricts topology to the prompt–response interface: TOHA’s divergence is a first-order attention statistic plus a computable correction, and on real heads the correction adds at most 0.007 AUC.

Contributions.

1. **A whole-barcode sink reduction** (§3). The ingredients are classical: a flag complex with a dominating vertex is a cone (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and persistence is stable (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). We combine them into one-sided bounds in a single computable number. For any symmetric dissimilarity and vertex s , bars of dimension $k \geq 1$ have length at most δ_s , and 0D total persistence is sandwiched around the star weight (Theorem 1). The whole barcode is within bottleneck distance δ_s of the star diagram, so every stable vectorization reduces to one column of distances (Corollary 1). For causal attention, δ_0 has a closed form (Corollary 2).
2. **The published topological detector is a first-order detector** (§3, §5.4). TOHA’s topological divergence equals one minus the response tokens’ mean max-prompt attention on prompt-coned heads, and is within a computable defect of it otherwise (Proposition 1). This makes exact the qualitative remark of Binkowski et al. (2026). On 18,906,944 real head graphs the identity has no violations, and adding TOHA to its first-order counterpart changes AUC by 0.000 to +0.007.
3. **Verification of the whole-graph reduction** (§5.1, Appendix D). There are zero bound violations in 838,356 graphs. Graphs are exactly coned in 36–98% of cases in sink-dominated models, H_1 is empty on every coned graph, and a sink dose-response experiment traces the weakness of 1D features to coning.
4. **A detection audit with strong baselines** (§5.3). On TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark, we compare 0D/1D PH, sink features, sink-deflated PH, the coning defect, and TOHA against length, lexical, log-probability, LLM-Check (Sriramanan et al., 2024), Lookback-Lens (Chuang et al., 2024), and hidden-state probes, with grouped cross-validation, early- and late-fusion incremental tests, and equivalence tests.

2 BACKGROUND AND RELATED WORK

Topology of attention. Kushnareva et al. (2021) and Cherniavskii et al. (2022) introduced barcodes of symmetrized BERT attention graphs as features for artificial-text detection and acceptability; Perez & Reinauer (2022) and Kostenok et al. (2023) extended them to classification and uncertainty estimation. For hallucination detection, TOHA (Bazarova et al., 2026) computes an MST-based divergence between prompt and response subgraphs, HalluZig (Samaga et al., 2026) uses zigzag persistence across layers, and CHARM (Frasca et al., 2026) learns a message-passing model over attention graphs. Gardinazzi et al. (2025) study zigzag persistence of hidden representations. All 0D summaries in this family are *MST* statistics (Fact 1; Gower & Ross, 1969); our results concern what those statistics measure in *causal* models.

Attention sinks. Xiao et al. (2024) observed that streaming LMs keep the first tokens’ key-values because a large share of attention lands there; Sun et al. (2024) tied sinks to massive activations, Cancedda (2024) to low-rank “dark signals”, Gu et al. (2025) characterized when they emerge during pre-training, and Barbero et al. (2025) argued they prevent over-mixing; related “no-op” attention to special tokens was observed in BERT (Clark et al., 2019) and vision transformers (Darcet et al., 2024). Binkowski et al. (2026) use sinks with large-norm values directly as a hallucination signal (SinkProbe) and argue that earlier attention-based detectors rely on them implicitly. Our theorem makes this precise for topological detectors, turning sink strength into a statement about the topology of attention graphs.

Hallucination detection. Beyond topology, detectors use sampling consistency (Manakul et al., 2023; Farquhar et al., 2024), internal probes (Azaria & Mitchell, 2023; Burns et al., 2022; Marks & Tegmark, 2023; Li et al., 2023b; Kossen et al., 2024; Orgad et al., 2024; Du et al., 2024), covariance spectra (Chen et al., 2024), attention-kernel log-determinants (Sriramanan et al., 2024), and context-vs-generated attention ratios (Chuang et al., 2024). Recent audits find that several benchmarks admit near-perfect detection from surface form alone (Hussain & Kantarcioglu, 2026; Janiak et al., 2025),

echoing annotation artifacts in NLI (Gururangan et al., 2018; McCoy et al., 2019); we therefore include lexical controls and an on-policy benchmark.

3 THEORY: ATTENTION SINKS CONE THE VIETORIS–RIPS FILTRATION

Let $D \in \mathbb{R}_{\geq 0}^{N \times N}$ be symmetric with zero diagonal. The Vietoris–Rips filtration is $K_\epsilon = \{\sigma \subseteq [N] : \max_{u,v \in \sigma} D_{uv} \leq \epsilon\}$; $\text{dgm}_k(D)$ is the barcode of $H_k(K_\epsilon; \mathbb{F})$ over a field, and $P_k(D)$ is its total persistence (sum of finite bar lengths). For attention, $A^{(l)}$ is the head-averaged, row-stochastic, lower-triangular attention matrix of layer l and, following the literature, $D = 1 - \max(A, A^\top)$ with zero diagonal.

Fact 1 (Gower & Ross, 1969). *All 0D bars are born at 0, and their finite death times are, as a multiset, the edge weights of a minimum spanning tree (MST) of D . In particular $P_0(D) = w(\text{MST}(D))$.*

Definition 1 (Coning defect, star weight). For $s \in [N]$, the *coning defect* is $\delta_s(D) = \max_{u \neq v, u, v \neq s} [\max(D_{su}, D_{sv}) - D_{uv}]_+$ (zero if $N < 3$), and the *star weight* is $S_s(D) = \sum_{u \neq s} D_{su}$. We call D *coned at s* if $\delta_s(D) = 0$, i.e. s is at least as close to both endpoints of every edge as they are to each other.

Theorem 1 (Sink-coning reduction). *For every symmetric D and every vertex s , with $\delta = \delta_s(D)$:*

- (a) *every bar of $\text{dgm}_k(D)$, $k \geq 1$, has length at most δ ; hence $P_k(D) \leq \delta \cdot |\text{dgm}_k(D)|$;*
- (b) *$S_s(D) - (N-1)\delta \leq P_0(D) \leq S_s(D)$, and $\max_u D_{su} - \delta \leq \max \text{dgm}_0^{\text{fin}}(D) \leq \max_u D_{su}$;*
- (c) *if D is coned at s , then $H_k(K_\epsilon) = 0$ for all $k \geq 1$ and all ϵ , the star at s is an MST, and $P_0(D) = S_s(D)$.*

Proof sketch (full proof in Appendix A). Let $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$ and $D'_{su} = D_{su}$. Then $D \leq D' \leq D + \delta$ entrywise and D' is coned at s . Since D' is coned, any simplex $\sigma \not\ni s$ with $|\sigma| \geq 2$ has the same filtration value as $\sigma \cup \{s\}$, and a vertex not yet joined to s has no edges; so each K'_ϵ is a cone with apex s plus isolated points, and $H_k(K'_\epsilon) = 0$ for $k \geq 1$. The inclusions $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$ factor the persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ through zero, so no bar outlives δ . For (b), the star is a spanning tree of D (upper bound) and an MST of D' ; MST weight and bottleneck are $(N-1)$ - and 1-Lipschitz in the sup-norm. The ingredients are standard: Lemma 1 is the dominating-vertex (cone) argument behind strong and edge collapses (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and the last step is a one-sided form of the interleaving argument behind persistence stability (Chazal et al., 2009). $\square$

The bound in (a) is attained: if $D_{su} = 1$ for all u and four further vertices form a 4-cycle with side $\frac{1}{2}$ and diagonals 1, then $\delta_s = \frac{1}{2}$ and H_1 has the single bar $[\frac{1}{2}, 1)$.

Theorem 1 controls summaries; the same comparison controls the whole barcode. Call $\Sigma_s(D)$ the *star diagram*: in degree 0 the bars $[0, D_{su})$, $u \neq s$, plus one infinite bar, and nothing in degree $k \geq 1$.

Corollary 1 (The whole barcode is the star, up to δ_s). *For every k , $d_B(\text{dgm}_k(D), \Sigma_s(D)_k) \leq \delta_s$, with equality of the diagrams when $\delta_s = 0$. More precisely, the j -th smallest finite 0D death of D lies in $[D_{s(j)} - \delta_s, D_{s(j)}]$, where $D_{s(j)}$ is the j -th smallest entry of $(D_{su})_{u \neq s}$, and β_0 of K_ϵ lies between $1 + \#\{u : D_{su} > \epsilon + \delta_s\}$ and $1 + \#\{u : D_{su} > \epsilon\}$. Hence every vectorization that is L -Lipschitz in d_B (e.g. persistence landscapes; Bubenik, 2015) is within $L\delta_s$ of a function of the column $(D_{su})_u$. Vectorizations that are stable in 1-Wasserstein distance (e.g. persistence images; Adams et al., 2017) are within $L((N-1)\delta_s + \frac{1}{2}\delta_s \sum_{k \geq 1} |\text{dgm}_k(D)|)$ of such a function. On coned graphs, every vectorization, including persistence entropy and Betti curves, is exactly such a function.*

Corollary 2 (Causal attention). *Let A be lower-triangular and row-stochastic, $D = 1 - \max(A, A^\top)$, and $s = 0$ the first token. Then $D_{0u} = 1 - A_{u0}$, so $S_0 = (N-1)(1 - \bar{m})$ with $\bar{m} = \frac{1}{N-1} \sum_{u \geq 1} A_{u0}$ the mean sink attention, and*

$$\delta_0 = \max_{1 \leq v < u} [A_{uv} - \min(A_{u0}, A_{v0})]_+.$$

Thus $P_0 \in [(N-1)(1 - \bar{m} - \delta_0), (N-1)(1 - \bar{m})]$ and all $H_{k \geq 1}$ bars have length at most δ_0 . The graph is exactly coned iff, for all $1 \leq v < u$, $A_{uv} \leq A_{u0}$ and $A_{uv} \leq A_{v0}$: no token attends to an earlier non-sink token more than either of the two attends to the sink.

Remark 1 (Choice of distance). Coning is an order condition, so it is invariant under any entrywise strictly increasing transform f of D with $f(0) = 0$ (e.g. $-\log$ of the symmetrized attention, or ranks without ties, in place of $1 - A$). The set of exactly coned graphs, and the vanishing of their higher homology, does not depend on how attention is turned into a distance; only the magnitude of δ_s on non-coned graphs does. (A merely non-decreasing f , such as clipping, can create coning.)

Corollary 2 makes four testable predictions: **(C1)** in coned layers 0D total persistence equals $(N-1)(1 - \bar{m})$, so after dividing by $N-1$ it is an affine function of sink attention; **(C2)** H_1 is empty whenever $\delta_0 = 0$; **(C3)** 1D persistence can only be non-zero where the sink is violated, with bars no longer than δ_0 ; **(C4)** removing the sink and renormalizing (*deflation*) exposes whatever non-sink topology exists.

Random weights and directed complexes. For i.i.d. weights, by contrast to Theorem 1, 1D total persistence grows linearly in N (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); we give an elementary explicit bound, $\mathbb{E}[P_1] \geq \frac{3}{8}N - \sqrt{3N} + O(1)$ (Proposition 2, Appendix A). For causal attention the directed flag complex is no escape: the digraph is a transitive tournament, so its weighted directed flag complex is isomorphic to $\text{VR}(D)$ as a filtered complex (Proposition 3, Appendix A.4).

The published topological detector reduces to a first-order statistic. TOHA (Bazarova et al., 2026) scores an attention head by the *topological divergence* between the response tokens R and the prompt tokens P . Its per-head score is $d(R, P) = \text{MTopDiv}(R, P)/|R|$, where MTopDiv is the total 0D persistence of the head’s graph $D = 1 - \max(A, A^\top)$ after all prompt–prompt distances are set to 0 (equivalently, the minimum spanning forest attaching R to P). TOHA averages d over at most ten heads chosen on labelled data. For $u \in R$ let $\pi_u = \max_{v \in P} A_{uv}$ be u ’s largest attention to any prompt token.

Proposition 1 (TOHA is max-prompt attention, up to a defect). *Let A be lower-triangular and row-stochastic, and let $P = \{0, \dots, p-1\}$ and $R = \{p, \dots, N-1\}$. With $\delta_P = \max_{w < u \in R} [A_{uw} - \min(\pi_u, \pi_w)]_+$,*

$$\frac{1}{|R|} \sum_{u \in R} (1 - \pi_u) - \delta_P \leq d(R, P) \leq \frac{1}{|R|} \sum_{u \in R} (1 - \pi_u),$$

with equality when $\delta_P = 0$. Moreover $1 - \pi_u \leq 1 - A_{u0}$, with equality iff the first token is among u ’s most-attended prompt tokens. On a head where this holds for every $u \in R$ and $\delta_P = 0$, $d(R, P) = 1 - \frac{1}{|R|} \sum_{u \in R} A_{u0}$, the response tokens’ mean sink attention.

Proof sketch. Zeroing the prompt block merges P at scale 0, so the remaining finite 0D bars are the MST weights of the quotient graph on $\{p^*\} \cup R$, where $D_{p^*u} = \min_{v \in P} D_{uv} = 1 - \pi_u$ (Fact 1). Apply Theorem 1(b) at the apex p^* ; as in Corollary 2, its coning defect is δ_P and its star weight is $\sum_u (1 - \pi_u)$. Divide by $|R|$. $\square$

Binkowski et al. (2026) note that sink nodes are “likely to appear in the MST” of MTopDiv . Proposition 1 makes this exact, and it shows where the topology can enter: only through δ_P , which is non-zero only when a response token attends to an earlier response token more than one of the two attends to its best prompt token. We test the proposition on real heads, and TOHA against its first-order counterparts, in §5.4.

4 EXPERIMENTAL SETUP

Models and benchmarks. We use Qwen2.5-1.5B/3B/7B-Instruct, Phi-3-mini-4k-instruct, TinyLlama-1.1B-Chat, SmolLM-1.7B-Instruct, and Mistral-7B-Instruct-v0.2: 7 open models spanning 1.1B–7B parameters, with additional Mistral-7B analysis in Appendix I. **TruthfulQA** (Lin et al., 2022): all 817 questions, each paired with its best answer and first incorrect answer (1,634 rows) under each model’s chat format (Appendix B). **HaluEval QA** (Li et al., 2023a): the first 1,000 (Qwen2.5-3B) or 500 (Qwen2.5-1.5B) knowledge-grounded questions with right and hallucinated answers. Both are *teacher-forced*: the model reads dataset-written answers. **TriviaQA on-policy**: 2,000 random validation questions (Joshi et al., 2017); each model answers greedily (short-phrase

Table 1: **Theorem 1 on real attention.** *cells*: (example, layer) graphs; *viol.*: violations of any bound in Theorem 1(a)–(b) with $s = 0$; *coned*: share with $\delta_0 = 0$ exactly; $H_1 = \emptyset$ | *coned*: share of coned graphs with empty H_1 (C2 predicts 100%). Last two columns: median over layers of the Spearman correlation across examples between P_0 and S_0 (ρ_{raw}), and between $P_0/(N-1)$ and $1 - \bar{m} = S_0/(N-1)$ ($\rho_{/N}$), which removes the shared dependence on sequence length.

Benchmark	Model	$\bar{N}$	graphs	viol.	sink attn.	coned	$H_1 = \emptyset$ coned	ρ_{raw}	$\rho_{/N}$
TruthfulQA	Qwen2.5-3B	44.2	58,824	0	0.50	62.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-1.5B	44.2	45,752	0	0.53	64.5%	100.0%	1.000	1.000
TruthfulQA	Phi-3-mini	29.0	52,288	0	0.69	88.4%	100.0%	1.000	1.000
TruthfulQA	TinyLlama-1.1B	42.3	35,948	0	0.53	36.2%	100.0%	1.000	0.982
TruthfulQA	SmolLM-1.7B	32.4	39,216	0	0.02	0.0%	–	0.967	0.412
TruthfulQA	Mistral-7B	33.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Mistral-7B (chat)	34.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-7B	44.2	45,752	0	0.01	0.0%	–	0.976	0.513
HaluEval	Qwen2.5-3B	126.9	72,000	0	0.42	52.7%	100.0%	1.000	1.000
HaluEval	Qwen2.5-1.5B	128.1	28,000	0	0.45	62.5%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-3B	68.6	72,000	0	0.48	63.0%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-1.5B	68.8	56,000	0	0.52	50.6%	100.0%	1.000	1.000
TriviaQA (on-policy)	Phi-3-mini	46.1	64,000	0	0.69	87.1%	100.0%	1.000	1.000
TriviaQA (on-policy)	TinyLlama-1.1B	72.0	44,000	0	0.50	36.4%	100.0%	1.000	0.991
TriviaQA (on-policy)	Mistral-7B	58.7	64,000	0	0.64	65.8%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-7B	68.3	56,000	0	0.01	0.0%	–	0.986	0.489

instruction, ≤ 24 tokens), an answer is labelled hallucinated unless a normalized gold alias occurs in it, and features are extracted by re-reading the model’s *own* answer. Per-model accuracies are in Table 6. All models run in bfloat16 with eager attention.

Features. Per layer (head-averaged A): 0D (max lifetime, total persistence); 1D (four summaries, divided by N); SINK (S_0 and $\max_u D_{0u}$, i.e. exactly the two quantities Corollary 2 says 0D reduces to); DEFLATED PH (0D+1D after deleting token 0 and renormalizing rows); ANSWER-ONLY PH (induced subgraph on answer tokens); row entropy and row max. Per head: 0D (MST total and max via a dense Prim’s algorithm), sink mass, entropy, Lookback ratio (Chuang et al., 2024), and the LLM-Check attention score (Sriramanan et al., 2024). TOHA (per head, separate forward pass, same rows): the exact score d of Bazarova et al. (2026) via a dense Prim’s algorithm on the prompt-contracted graph, the $\bar{\pi}$ -score, the response sink score, and δ_P (Proposition 1). Other: answer-token log-probability and predictive-entropy statistics; mean-pooled answer hidden states at the middle layer; TF-IDF word and character n -grams of the answer text (LEXICAL).

Protocol. Every number uses one evaluator: standardized features and L_2 logistic regression, with $C=1$ for banks of at most 64 dimensions and otherwise $C \in \{0.01, 0.1, 1\}$ chosen by inner grouped 3-fold CV. The outer loop is 10-fold StratifiedGroupKFold grouped by question, repeated over 3 seeds (42–44), with out-of-fold probabilities averaged. Incremental value is tested in two ways. *Early fusion* (“X|Y”) concatenates the banks. *Late fusion* (“X $\oplus$ Y”) stacks the two banks’ out-of-fold logits in a second-level logistic regression under the same CV, so that a small bank cannot be drowned out by a wide one. Differences in pooled AUC use a 10,000-resample question-level cluster bootstrap. We call two banks *equivalent* if the 90% interval lies inside ± 0.015 AUC (two one-sided tests, TOST, at $\alpha=0.05$), and we report TOST power. Paired benchmarks also report within-pair accuracy. The theory checks use every (example, layer) graph. Ripser (Bauer, 2021) stores filtration values in 32-bit floats, so bar-length and max-death bounds are checked at tolerance 10^{-5} , and the P_0 sandwich, a sum of up to N such values, at 10^{-3} .

5 RESULTS

5.1 THE REDUCTION HOLDS ON REAL ATTENTION

Table 1 checks Theorem 1 on every layer-level graph of every example: 838,356 graphs in total. There are 0 violations of any bound. In the 13 settings whose models place 0.42 to 0.75 of attention on token 0, 36–98% of graphs are *exactly* coned ($\delta_0 = 0$), and on every one of them H_1 is empty, as prediction C2 requires. Across graphs, δ_0 and 1D total persistence have Spearman correlation 0.07 to 0.89 (C3), the floor being the 2 settings above 90% coned, where both quantities are near-constant (99.99% of graphs have no 1D bars) and the correlation is degenerate rather than weak; elsewhere it

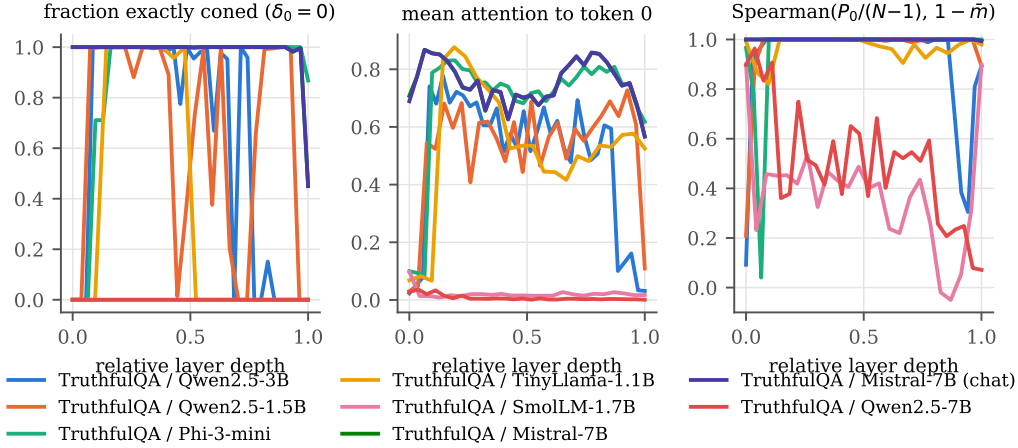


Figure 1: **Layerwise structure (TruthfulQA).** Coning (left) tracks sink attention (middle): layers with little sink attention, typically the first few and, in some models, the last few, are rarely coned, and the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B) are never coned at $s = 0$. Right: length-normalized rank correlation between P_0 and sink attention. It is near 1 in the layers of sink models, dips in some open layers, and is low throughout the two without a sink.

is at least 0.32. Both P_0 and the star weight $S_0 = (N-1)(1 - \bar{m})$ scale with sequence length, so their raw per-layer rank correlation (median ≥ 0.999) partly reflects length. After dividing both by $N-1$, the median per-layer rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ is still at least 0.98 in every sink setting (C1). In most layers, then, length-normalized 0D total persistence is, up to rank, one minus the mean attention paid to the first token. The weakest single layer per setting reaches only -0.14 to 0.99.

Figure 1 shows where coning happens: in the middle of the network, where sink attention is high. The first few layers are never sink-dominated, and in some models (TinyLlama, the Qwen models) so is a block of late layers; these layers are rarely coned. There, the length-normalized correlation can drop, as Theorem 1(b) allows once $(N-1)\delta_0$ is comparable to the spread of S_0 across examples.

Models without a first-token sink as a contrast. Our study includes 2 models that place almost no attention on token 0 — Qwen2.5-7B and SmolLM-1.7B, at 0.01 to 0.02 — and none of their graphs is coned at $s = 0$, so Theorem 1 at the first token gives no useful guarantee for them. They are not alike. Allowing any apex, 0% of SmolLM-1.7B’s graphs are coned — no vertex cones any of them — against 10% of Qwen2.5-7B’s, whose attention is strongly concentrated, just not on token 0 (§5.4): one model has no cone structure to find, the other is one where $s = 0$ is the wrong apex. Their H_1 is non-empty on 55% to 87% of graphs, and although the raw $\rho(P_0, S_0)$ is 0.967 to 0.986, this is almost entirely shared length: the length-normalized correlation is 0.41 to 0.51. Both differ from the sink-dominated models in weights, data and template, so this is a contrast, not a controlled intervention; the within-model intervention is deflation (C4, below). The reduction is a property of sink-dominated attention, not of the pipeline.

5.2 WHAT TOPOLOGY ADDS BEYOND THE SINK

If 0D persistence reduced *exactly* to the star, a classifier on the two sink features would match one on the two 0D features in every layer. Table 13 (“Sink vs. 0D”) shows the two banks differ by -0.015 to +0.031 AUC across the 13 sink settings evaluated, and are statistically equivalent within ± 0.015 in 8 of them (the TOST power of this comparison is modest, so non-equivalence is not evidence of a difference). Adding 0D to the sink bank gains 0.000 to +0.028.

The residual lives exactly where the theory says it can. We split layers without using labels: *coned* layers ($\geq 90\%$ of graphs coned) and *open* layers ($< 50\%$). Adding the coned layers’ 0D features to SINK changes AUC by -0.001 to +0.004, whereas the open layers account for +0.003 to

Table 2: **Pooled out-of-fold ROC-AUC** (3-seed repeated grouped 10-fold CV; seed $SD \leq 0.008$). TQA: TruthfulQA; HE: HaluEval; TrQA: on-policy TriviaQA. Lex.: answer-text TF-IDF; LogP: log-probability statistics; LLM-C.: LLM-Check; Sink: $(S_0, \max_u D_{0u})$ per layer; 0D/1D: persistence per layer; subscript h : per head (Sink $_h$: sink mass; 0D $_h$: MST total and max); Lookb.: Lookback ratios; Hidden: mid-layer probe; Non-topo.: LogP+LLM-C.+Lookb.+Hidden+row statistics. Best per row in bold. Full table in Appendix G.

Setting	Lex.	LogP	LLM-C.	Sink	0D	1D	Sink $_h$	0D $_h$	Lookb.	Hidden	Non-topo.
TQA Qw3B	.820	.630	.693	.737	.736	.660	.851	.853	.897	.913	.926
TQA Qw1.5B	.820	.638	.666	.723	.726	.586	.830	.838	.851	.916	.917
TQA Phi-3	.820	.578	.653	.651	.646	.526	.871	.868	.892	.933	.936
TQA TinyLL	.820	.637	.614	.643	.654	.592	.808	.806	.850	.883	.885
TQA SmolLM	.820	.588	.604	.611	.583	.543	.720	.800	.863	.894	.899
TQA Mis7B	.820	.641	.652	.685	.684	.501	.903	.893	.933	.956	.959
TQA Mis7B (chat)	.820	.643	.653	.688	.688	.501	.907	.900	.930	.958	.960
TQA Qw7B	.820	.674	.667	.749	.679	.610	.821	.850	.922	.933	.950
HE Qw3B	.981	.993	.745	.829	.811	.710	.958	.971	.998	1.00	1.00
HE Qw1.5B	.979	.995	.791	.820	.789	.747	.922	.949	.999	1.00	1.00
TrQA Qw3B	.596	.839	.688	.717	.732	.601	.820	.807	.861	.841	.870
TrQA Qw1.5B	.609	.823	.676	.707	.716	.574	.767	.768	.824	.855	.873
TrQA Phi-3	.635	.877	.676	.707	.710	.546	.826	.807	.840	.867	.883
TrQA TinyLL	.740	.724	.688	.720	.720	.659	.814	.775	.823	.827	.842
TrQA Mis7B	.677	.719	.686	.696	.699	.611	.812	.775	.822	.833	.841
TrQA Qw7B	.594	.847	.725	.714	.668	.582	.801	.798	.879	.854	.881

+0.027 (Appendix G). Almost everything 0D adds beyond sink attention comes from the layers where the sink does not dominate.

Per head, the reduction is tighter. Per-head sink mass and per-head 0D are far stronger than their head-averaged versions (Table 2), but adding per-head MST total ($P_{0,h}$) to per-head sink mass changes AUC by only -0.001 to +0.011, which is equivalent within ± 0.015 in 13 of the 13 sink settings with per-head features.

The coning defect is a signal, but not a new one. Theorem 1 singles out δ_0 as the only place where topology can depart from the sink, so we also test it as a detector. Head-averaged, it is weak (AUC 0.50 to 0.67). Per head it is strong (0.69 to 0.96), and stacked onto per-head sink mass it adds +0.004 to +0.029 AUC (95% interval above 0 in 16 of 16 settings). Stacked onto the non-topological probes, however, it adds only 0.000 to +0.009 (above 0 in 5, equivalent within ± 0.015 in 16 of 16).

Deleting the sink exposes some, redundant, structure. Deflated PH (C4) adds +0.029 to +0.067 over SINK on TruthfulQA (95% interval excludes 0 in 6 of 6 sink settings) but only +0.002 to +0.043 on the on-policy benchmark (2 of 5). Answer-only PH behaves similarly (Appendix G). So attention graphs carry some non-sink geometry, most clearly on teacher-forced text. As §5.3 shows, this geometry is already available to non-topological probes.

One-dimensional persistence. Most real graphs have no 1D bars at all (Table 1), unlike i.i.d. weights, where 1D persistence grows linearly. Adding 1D to 0D changes AUC by -0.008 to +0.033 and is equivalent within ± 0.015 in 9 of 13 sink settings. A sink dose-response experiment and a planted-cycle experiment (Appendix D) show why: under a strong sink, a loop is visible to H_1 only if it breaks the cone.

5.3 HALLUCINATION DETECTION AGAINST STRONG BASELINES

Table 2 places attention-graph topology among standard detectors. Three patterns hold across benchmarks.

Topology is dominated by cheap non-topological probes. On TruthfulQA and the on-policy benchmark, a mean-pooled hidden-state probe (0.83 to 0.96) and per-head Lookback ratios (0.82 to 0.93) outperform every topological bank, and their combination with log-probability, LLM-Check, and row

Table 3: **TOHA against its first-order counterparts** (Proposition 1). $\delta_P=0$: share of (example, head) graphs that are exactly prompt-coned, on which d equals the $\bar{\pi}$ -score. $\rho(d, \bar{\pi})$: median over heads of the rank correlation across examples. AUC columns: TOHA’s head-selection algorithm run on d (TOHA), on the $\bar{\pi}$ -score (TOHA $[\bar{\pi}]$), and on the response sink score (TOHA $[\bar{a}_0]$); then supervised probes on all heads of each score; then the non-topological bank. Δ columns and marks as in Table 13; $\oplus$ is late fusion.

Setting	$\delta_P=0$	$\rho(d, \bar{\pi})$	TOHA	TOHA $[\bar{\pi}]$	TOHA $[\bar{a}_0]$	TOHA $_{\text{sup}}$	$\bar{\pi}_{\text{sup}}$	$\bar{a}_0_{\text{sup}}$	Non-topo.	$\bar{\pi}$ -TOHA	$\bar{a}_0$ -TOHA	TOHA $\oplus\bar{\pi}$	TOHA $\oplus\text{NT}$
TQA Qw3B	57%	.976	.668	.670	.649	.892	.892	.892	.926	+0.003 $\equiv$	-0.018*	+0.003 $\equiv$	+0.003 $\equiv$
TQA Qw1.5B	58%	.978	.646	.649	.628	.858	.857	.857	.917	+0.003 $\equiv$	-0.018*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Phi-3	74%	.996	.751	.742	.715	.898	.901	.905	.936	-0.009*	-0.036*	0.000 $\equiv$	0.000 $\equiv$
TQA TinyLL	74%	.995	.666	.655	.618	.845	.843	.848	.885	-0.012*	-0.049*	+0.002 $\equiv$	+0.001 $\equiv$
TQA SmoLLM	58%	.954	.701	.695	.595	.862	.860	.841	.899	-0.007 $\equiv$	-0.107*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Mis7B	81%	.998	.751	.764	.766	.932	.934	.936	.959	+0.013*	+0.015*	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw7B	57%	.980	.705	.709	.544	.925	.927	.906	.950	+0.004 $\equiv$	-0.161*	+0.001 $\equiv$	+0.004 $\equiv$
HE Qw3B	68%	.987	.967	.982	.961	.999	.999	1.00	1.00	+0.015*	-0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	69%	.988	.970	.988	.972	.999	.999	.999	1.00	+0.018*	+0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	76%	.991	.728	.736	.742	.862	.863	.870	.870	+0.009	+0.014*	+0.002 $\equiv$	+0.012*
TrQA Qw1.5B	77%	.992	.712	.714	.708	.828	.827	.836	.873	+0.002 $\equiv$	-0.005	+0.003 $\equiv$	+0.004 $\equiv$
TrQA Phi-3	88%	.998	.724	.731	.733	.847	.851	.850	.883	+0.007 $\equiv$	+0.008	0.000 $\equiv$	+0.001 $\equiv$
TrQA TinyLL	66%	.994	.714	.725	.693	.833	.826	.832	.842	+0.011*	-0.021*	+0.007 $\equiv$	+0.010*
TrQA Mis7B	75%	.993	.716	.706	.710	.818	.820	.817	.841	-0.010	-0.007	+0.002 $\equiv$	+0.008 $\equiv$
TrQA Qw7B	76%	.994	.754	.790	.617	.876	.880	.842	.881	+0.036*	-0.137*	+0.001 $\equiv$	+0.012*

statistics reaches 0.84 to 0.96. Adding 0D PH to that combination changes AUC by 0.000 to +0.001 (equivalent within ± 0.015 in 16 of the 16 settings with per-head features), and adding sink-deflated PH changes it by -0.001 to +0.002 (equivalent in 16 of 16). Early fusion could hide a small bank inside a $\sim 2,000$ -dimensional one, so we repeat the test with late fusion. Stacking the 0D score onto the non-topological score changes AUC by 0.000 to +0.003 (equivalent in 16 of 16 settings; the 95% interval lies above 0 in 0 of them), and stacking the deflated-PH score changes it by 0.000 to +0.006 (equivalent in 16 of 16).

Which cheap probe wins depends on whether text is teacher-forced. On TruthfulQA, token log-probability is weak (0.58 to 0.67), consistent with the benchmark’s design around imitative falsehoods, and 0D PH (0.58 to 0.74) scores higher for 7 of 8 models, significantly so for 5. On-policy, log-probability is strong (0.72 to 0.88) and significantly beats 0D (0.67 to 0.73) for 4 of 6 models. Conclusions about topology versus confidence drawn from a single teacher-forced benchmark do not transfer.

Per-head resolution matters more than topology. Per-head features of every kind are far stronger than head-averaged ones, and per-head entropy lands within -0.016 to +0.044 AUC of per-head 0D (Appendix G).

5.4 THE PUBLISHED TOPOLOGICAL DETECTOR IS FIRST-ORDER

We test Proposition 1 on every head of every example: 18,906,944 head graphs from 15 settings (Table 3). We then run TOHA’s head-selection algorithm (Bazarova et al., 2026, Alg. 1) inside our cross-validation, selecting heads on the whole training fold, which is larger than TOHA’s probe set. We run it once on TOHA’s score d and, unchanged, on the two first-order scores the proposition names: the $\bar{\pi}$ -score $\frac{1}{|R|} \sum_u (1 - \pi_u)$ and the response sink score $1 - \bar{a}_0 = \frac{1}{|R|} \sum_u (1 - A_{u0})$.

The identity holds on real heads. The sandwich has 0 violations. The score d is our own implementation, so we also check it against the authors’ released code: on 21,696 head graphs from 2 settings, their MTOP-DIV routine (Bazarova et al., 2026), which uses ripser rather than our dense Prim’s algorithm, agrees with ours to 1.2×10^{-8} . Per setting, 57–88% of head graphs are exactly prompt-coned, and on those d equals the $\bar{\pi}$ -score up to float rounding (largest gap $6 \cdot 10^{-8}$). Across examples, the median per-head rank correlation between d and the $\bar{\pi}$ -score is 0.954 to 0.998. With the sink score it is 0.78 to 0.99 in the sink models, but -0.16 to -0.09 in Qwen2.5-7B and SmoLLM-1.7B, which have no first-token sink: there d tracks the largest prompt attention just as tightly while being, if anything, mildly anti-correlated with attention to token 0. The reduction is to max-prompt attention; that this coincides with the sink is a property of the model, not of the theorem. Shifting the sink’s attention logit moves these quantities as predicted (Appendix H).

Topology adds almost nothing to the first-order score. Stacking d onto the $\bar{\pi}$ -score changes AUC by 0.000 to +0.007. The change is equivalent to zero within ± 0.015 in 15 of 15 settings, although the

95% interval lies above 0 in 4 of them. With supervised probes on all heads (the setting of TOHA’s own feature comparison), the $\bar{\pi}$ -score differs from d by -0.007 to +0.004 (equivalent in 15 of 15). The response sink score differs from d by -0.034 to +0.008 (equivalent in 11). The exceptions are the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B), where the sink score is worse, and one on-policy setting whose interval extends past the margin on the positive side. TOHA’s discrete head selection is noisier: run on the $\bar{\pi}$ -score it changes AUC by -0.012 to +0.036 (significantly higher in 5 settings, lower in 2, non-inferior at margin 0.015 in 12 of 15). Run on the sink score it loses up to 0.049 AUC in the sink models (0.107 to 0.161 in Qwen2.5-7B and SmolLM-1.7B, which have no first-token sink): the selected heads are prompt-coned on 57–95% of examples, but the sink is every response token’s top prompt token on only 19–79%. Stacked onto the non-topological probes, TOHA changes AUC by 0.000 to +0.012 (95% interval above 0 in 6 of 15); by the above, whatever it adds is carried by the $\bar{\pi}$ -score.

Why “attention to $\langle s \rangle$ ” looked weak. Bazarova et al. (2026, Table 9) report that attention to the first token is a much weaker feature than their divergence. Proposition 1 identifies the quantity that d actually tracks: per head, the attention of the *response* tokens to their most-attended prompt token. In the sink models, that token is the sink for every response token in 40–83% of head graphs.

Benchmark artifacts. A TF-IDF model on the answer text alone reaches 0.82 AUC on TruthfulQA and 0.98 on HaluEval, but only 0.59 to 0.74 on-policy: teacher-forced benchmarks reward recognizing how the dataset’s answers were *written* (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). We draw no conclusions from HaluEval, where lexical separability is near perfect.

6 DISCUSSION

What attention-graph topology measures. In decoder-only models, Vietoris–Rips persistence of an attention graph, and every stable summary of it, is controlled by one scalar per graph: the coning defect of the first token. The same is true of TOHA’s prompt–response divergence, with the prompt playing the role of the apex. Where it is zero, as in most middle-layer graphs of every sink-dominated model we tested, 0D total persistence equals $(N-1)(1 - \bar{m})$, a function of length and sink attention alone, and higher homology is empty. Where it is positive, Theorem 1 bounds how far topology can depart from the sink; in our data this happens mainly in the first layers. Practical recommendations follow from this in Appendix E.

Limitations. Our models range over 1.1B–7B parameters; sink strength changes with scale and training (Gu et al., 2025), so the *share* of coned layers is model-specific, although Theorem 1 is not. Our results cover Vietoris–Rips persistence of symmetrized attention (head-averaged or per head), its causal directed-flag equivalent, every stable vectorization of it, and TOHA. They do not cover HalluZig (Samaga et al., 2026), which computes zigzag persistence of the *graph* (1-skeleton) homology of top-percentile edge sets across layers: a cone graph still has cycles, because its triangles are not filled, so Theorem 1 does not constrain its H_1 . Learned graph models (Frasca et al., 2026) are also outside the theorem, although their inputs are the same sink-dominated graphs. TOHA was designed for RAG; only HaluEval supplies a context passage here. Our on-policy labels come from alias string matching, which mislabels some paraphrases and hedged answers. An LLM-judge audit of 200 answers in 2 on-policy settings finds 6% to 14% disagreement; re-scoring those rows under the judge’s labels shifts every bank by at most 0.058 AUC and leaves the dominance orderings intact. The judge is itself a model, so the labels remain unaudited against human judgment (Janiak et al., 2025). TinyLlama’s on-policy answers were generated without the BOS token added at feature extraction. Our intervals resample questions for a fixed, seed-averaged predictor and do not include variation from refitting, so equivalence claims are conditional on the trained models. Prompt formats differ across models (Appendix B), so cross-model comparisons mix architecture and template.

REPRODUCIBILITY STATEMENT

All code is released at <https://anonymous.4open.science/r/ICLR-BE07/.sinktda/extract.py> computes all whole-graph and per-head features, `sinktda/toha.py` computes TOHA and its first-order counterparts and runs TOHA’s head selection, and `sinktda/evaluate.py` produces every AUC, interval and theory check.

sinktda/late_fusion.py and sinktda/defect_probe.py run the stacked tests, sinktda/check_theory.py and sinktda/synthetic.py run the numerical and synthetic checks, and sinktda/report.py generates every table, figure and number in the text. The run order is in sinktda/run_fix.sh and sinktda/run_toha.sh. Per-example features are included for 5 of the 16 settings (TQA Mis7B, TQA Mis7B (chat), TQA Qw7B, TrQA Mis7B and TrQA Qw7B), and the on-policy generations for TrQA Mis7B and TrQA Qw7B. The remaining feature dumps were lost with the machine that produced them; sinktda/extract.py regenerates them from the model and setting identifiers in Appendix B. Seeds, prompts, dtype and model identifiers are listed in Appendix B; proofs are in Appendix A.

LLM USAGE STATEMENT

LLMs were used to help implement experiment code, check reported statistics against result files, and draft and edit text. The authors verified every claim, number, proof, and citation and take full responsibility for the paper.

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A PROOFS

Throughout, D is a symmetric $N \times N$ matrix with non-negative entries and zero diagonal, K_ϵ is the Vietoris–Rips complex at scale ϵ , $\text{filt}_D(\sigma) = \max_{u,v \in \sigma} D_{uv}$ (with $\text{filt}_D(\{u\}) = 0$), and homology is taken over a field. Bars are half-open intervals $[b, d)$.

A.1 FACT 1

All vertices enter at 0. Processing edges in non-decreasing weight order, an edge either joins two components, in which case (elder rule) one 0D class dies at its weight and Kruskal’s algorithm adds the edge to the spanning forest, or it closes a cycle, in which case no 0D class dies and Kruskal discards it. Hence the $N - 1$ finite death times are the MST edge weights (Gower & Ross, 1969; Edelsbrunner & Harer, 2010).

A.2 THEOREM 1

Lemma 1 (Coned filtrations). *If D is coned at s , then for every $\epsilon \geq 0$, $K_\epsilon = C_\epsilon \sqcup I_\epsilon$, where C_ϵ is a simplicial cone with apex s and I_ϵ is a set of isolated vertices. Consequently $\tilde{H}_k(K_\epsilon) = 0$ for every $k \geq 1$.*

Proof. Let $\sigma \in K_\epsilon$ with $s \notin \sigma$ and $|\sigma| \geq 2$. For each $u \in \sigma$ choose $v \in \sigma \setminus \{u\}$; coning gives $D_{su} \leq D_{uv} \leq \text{filt}_D(\sigma) \leq \epsilon$. Therefore $\text{filt}_D(\sigma \cup \{s\}) = \text{filt}_D(\sigma)$ and $\sigma \cup \{s\} \in K_\epsilon$. Now let $u \neq s$ be a vertex with $D_{su} > \epsilon$. If u had an edge $\{u, v\} \in K_\epsilon$ then $D_{su} \leq D_{uv} \leq \epsilon$, a contradiction; so u is isolated. Let I_ϵ be these vertices and C_ϵ the subcomplex on the remaining vertices. Every vertex of C_ϵ is adjacent to s , and by the first part every simplex of C_ϵ extends by s , so C_ϵ is the cone over its link of s and is contractible. H_k of a disjoint union is the direct sum, and isolated points have no homology in degree ≥ 1 . $\square$

Proof of Theorem 1. Fix s and $\delta = \delta_s(D)$. Define $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$, $u \neq v$; $D'_{su} = D'_{us} = D_{su}$; $D'_{uu} = 0$.

Step 1: $D \leq D' \leq D + \delta$ and D' is coned at s . The first inequality is immediate. For $u, v \neq s$, $D'_{uv} - D_{uv} = [\max(D_{su}, D_{sv}) - D_{uv}]_+ \leq \delta$. Coning: $\max(D'_{su}, D'_{sv}) = \max(D_{su}, D_{sv}) \leq D'_{uv}$.

Step 2: interleaving. Since filtration values are maxima of entries, $\text{filt}_D \leq \text{filt}_{D'} \leq \text{filt}_D + \delta$. Hence for every b , $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$, where K' is the filtration of D' .

Step 3: (a). The persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ is induced by the composite inclusion in Step 2 and therefore factors through $H_k(K'_{b+\delta})$, which is zero for $k \geq 1$ by Lemma 1. The rank of $H_k(K_b) \rightarrow H_k(K_{b'})$ equals the number of bars $[b_i, d_i)$ with $b_i \leq b$ and $d_i > b'$. A bar with $d_i - b_i > \delta$ would contribute to this rank at $b = b_i$, $b' = b_i + \delta$, a contradiction. So every bar has length at most δ , and summing gives the bound on P_k .

Step 4: (b). By Fact 1, $P_0(D) = w_D(T_D)$ for an MST T_D of D . The star $T_s = \{\{s, u\}\}_{u \neq s}$ is a spanning tree, so $P_0(D) \leq w_D(T_s) = S_s(D)$. For the lower bound, T_s is an MST of D' , even with ties. Root any spanning tree T at s and let e_u be the edge from $u \neq s$ to its parent $p(u)$. If $p(u) = s$ then $D'_{e_u} = D'_{su}$; otherwise $D'_{e_u} = D'_{up(u)} \geq \max(D'_{su}, D'_{sp(u)}) \geq D'_{su}$ because D' is coned. The map $u \mapsto e_u$ is a bijection onto the edges of T , so $w_{D'}(T) \geq \sum_{u \neq s} D'_{su} = w_{D'}(T_s)$ and $\max_{e \in T} D'_e \geq \max_u D'_{su}$. Hence $w_{D'}(T_{D'}) = w_{D'}(T_s) = S_s(D)$, and $P_0(D) = w_D(T_D) \geq w_{D'}(T_D) - (N-1)\delta \geq w_{D'}(T_{D'}) - (N-1)\delta = S_s(D) - (N-1)\delta$. The maximum finite 0D death is the bottleneck value $\min_T \max_{e \in T} w(e)$, which is attained by any MST; it is at most $\max_u D_{su}$ (the star is a spanning tree) and is 1-Lipschitz in the sup-norm, which gives the lower bound because the bottleneck of D' is $\max_u D_{su}$ (by the rooted-tree argument above).

Relation to known results. Lemma 1 is the classical observation that a flag complex in which one vertex is adjacent to every non-isolated vertex, and to every simplex, is a cone; this is the dominated-vertex mechanism behind strong collapses (Barmak & Minian, 2012) and edge collapses of flag filtrations (Boissonnat & Pritam, 2020). Step 3 is a one-sided instance of the interleaving argument behind the stability of persistence (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). The

two-sided stability theorem would give bars of length at most 2δ , whereas the one-sided comparison $D \leq D'$ gives δ , which is attained (Section 3).

Step 5: (c). If $\delta = 0$ then $D' = D$, and the claims follow from Lemma 1 and Step 4. $\square$

Proof of Corollary 1. Let D' be as in the proof of Theorem 1. Since D' is coned at s , Lemma 1 and Step 4 give $\text{dgm}_k(D') = \Sigma_s(D)_k$ for all k : no bars in degree $k \geq 1$, and 0D deaths equal to the star weights $D'_{su} = D_{su}$. For $1 \leq j \leq N-1$, the j -th smallest finite 0D death of a matrix E is $w_j(E) = \min\{\epsilon : \beta_0(K_\epsilon^E) \leq N-j\}$. If $E \leq E'$ entrywise then $K_\epsilon^{E'} \subseteq K_\epsilon^E$ on the same vertex set, so $\beta_0(K_\epsilon^E) \leq \beta_0(K_\epsilon^{E'})$ and $w_j(E) \leq w_j(E')$; and $w_j(E+c) = w_j(E) + c$ on off-diagonal entries. From $D \leq D' \leq D + \delta_s$ we get $w_j(D) \leq D_{s(j)} \leq w_j(D) + \delta_s$. The same inclusions give the Betti-0 sandwich, because $\beta_0(K_\epsilon^D) = 1 + \#\{u : D_{su} > \epsilon\}$. Matching the j -th deaths shows that $d_B(\text{dgm}_0(D), \Sigma_s(D)_0) \leq \delta_s$ and $W_1 \leq (N-1)\delta_s$. In degree $k \geq 1$, every bar has length at most δ_s (Theorem 1(a)), so matching it to the diagonal costs at most $\delta_s/2$ in ℓ^∞ . The Lipschitz statements follow from the stability of landscapes in d_B (Bubenik, 2015) and of persistence images in W_1 (Adams et al., 2017). If $\delta_s = 0$ then $D = D'$. $\square$

Proof of Proposition 1. Let $\tilde{D}$ equal D except that $\tilde{D}_{vv'} = 0$ for $v, v' \in P$. At scale 0, Kruskal's algorithm joins all of P into one component with $|P| - 1$ zero-length bars. Every later merge is a merge in the quotient graph Q on $\{p^*\} \cup R$ with $Q_{p^*u} = \min_{v \in P} \tilde{D}_{vu}$ and $Q_{uw} = D_{uw}$. So $\text{MTopDiv}(R, P) = P_0(Q)$. For $u \in R$ and $v \in P$ we have $v < u$ and $D_{uv} = 1 - A_{uv}$, so $Q_{p^*u} = 1 - \pi_u$. For $w < u$ in R , $Q_{uw} = 1 - A_{uw}$, and $\max(Q_{p^*u}, Q_{p^*w}) - Q_{uw} = A_{uw} - \min(\pi_u, \pi_w)$, so $\delta_{p^*}(Q) = \delta_P$. Theorem 1(b) with $N-1 = |R|$ gives the sandwich, with equality when $\delta_P = 0$ by part (c). Finally $\pi_u \geq A_{u0}$ because $0 \in P$. $\square$

We checked both statements numerically. The fast MTop-Div (a dense Prim's algorithm on Q) agrees with ripser run on $\tilde{D}$ to $3 \cdot 10^{-8}$ on 400 random causal matrices. On these, and on 300 further random matrices checked for the elementwise death bound and the bar bound in degrees 1 and 2, there are 0 violations (sinktda/check_theory.py).

Proof of Corollary 2. For $u > v$, $A_{vu} = 0$, so $D_{uv} = 1 - A_{uv}$; in particular $D_{0u} = 1 - A_{u0}$. For $1 \leq v < u$, $\max(D_{0u}, D_{0v}) - D_{uv} = A_{uv} - \min(A_{u0}, A_{v0})$. Substitute into Theorem 1. $\square$

A.3 LINEAR GROWTH OF 1D PERSISTENCE UNDER I.I.D. WEIGHTS

Length normalization of 1D features is usually motivated by the growth of cycles with N . For i.i.d. weights this growth is known: 0D total persistence converges to $\zeta(3)$ (Frieze, 1985), and 1D total persistence of the random clique-complex process is $\Theta(N)$ (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019). The following elementary bound makes the constant explicit.

Proposition 2 (Linear growth under i.i.d. weights). *If $\{D_{uv}\}_{u < v}$ are i.i.d. Uniform $[0, 1]$, then $\mathbb{E}[P_1(N)] \geq \int_0^1 \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+ dp = \frac{3}{8}N - \sqrt{3N} + O(1)$. (A matching $O(N)$ upper bound is Theorem 1.5 of Hino & Kanazawa, 2019.)*

Proof. With i.i.d. uniform weights, the edges with $D_{uv} \leq p$ form an Erdős-Rényi graph $G(N, p)$ and K_p is its clique (flag) complex. Every 1D bar is contained in $[0, 1]$, because K_1 is the full simplex. Writing $\beta_1(p)$ for the first Betti number of K_p , the number of bars alive at p is $\beta_1(p)$, so $P_1 = \int_0^1 \beta_1(p) dp$. For any finite complex with f_i faces of dimension i , the strong Morse inequality at degree 2 reads $\beta_2 - \beta_1 + \beta_0 \leq f_2 - f_1 + f_0$ (Edelsbrunner & Harer, 2010), hence $\beta_1 \geq f_1 - f_0 - f_2 + \beta_0 + \beta_2 \geq f_1 - f_0 - f_2$. Since also $\beta_1 \geq 0$, Jensen's inequality gives $\mathbb{E}\beta_1(p) \geq [\mathbb{E}f_1 - f_0 - \mathbb{E}f_2]_+ = \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+$. Integrating over p gives the first inequality. With $p = x/\sqrt{N}$ and $dp = dx/\sqrt{N}$, the integrand divided by $\sqrt{N}$ is $N(x/2 - x^3/6) - \sqrt{N} + O(1)$ uniformly for $x \in [0, \sqrt{3} + 1]$. It is positive on an interval whose endpoints are $O(N^{-1/2})$ from 0 and $\sqrt{3}$, and it is negative for larger x . Hence $\int_0^1 [\cdot]_+ dp = N \int_0^{\sqrt{3}} \left(\frac{x}{2} - \frac{x^3}{6} \right) dx - \sqrt{3N} + O(1) = \frac{3}{8}N - \sqrt{3N} + O(1)$. Numerically, $(\frac{3}{8}N - \text{bound})/\sqrt{N} = 1.62, 1.67, 1.70, 1.72$ at $N = 2^8, 2^{10}, 2^{12}, 2^{14}$,

approaching $\sqrt{3} \approx 1.73$. The order $\Theta(N)$ itself is known (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); the proposition only makes the constant explicit. $\square$

Numerically, the integral equals 0.791, 3.92, 11.9, 30.2, 70.1 at $N = 16, 32, 64, 128, 256$, against simulated means 1.91, 5.83, 15.5, 36.2, 79.3 under i.i.d. Uniform $[0, 1]$ weights (Figure 4, left). Over $N \in [16, 256]$ both grow faster than linearly (log-log slopes 1.55 and 1.31), a pre-asymptotic effect of the $-\sqrt{3N}$ term; both are $\Theta(N)$ as $N \rightarrow \infty$.

A.4 DIRECTED FLAG COMPLEXES ON CAUSAL ATTENTION

Proposition 3 (Directed flag collapse). *Let A be lower-triangular with non-negative entries (self-loops are ignored). Let $\text{DFl}(A)$ be the filtered directed flag complex (Lütgehetmann et al., 2020) of the digraph with an edge $i \rightarrow j$ for every $j < i$ (a transitive tournament) and weight $1 - A_{ij}$, and let $\text{VR}(D)$ be as above with $D = 1 - \max(A, A^\top)$. Then $\text{DFl}(A)$ and $\text{VR}(D)$ are isomorphic filtered complexes, so $\text{dgm}_k(\text{DFl}(A)) = \text{dgm}_k(\text{VR}(D))$ for all k . The same holds for any weighting $w(i \rightarrow j) = \phi(A_{ij})$ with $D_{ij} = \phi(\max(A_{ij}, A_{ji}))$ and ϕ non-increasing.*

Proof. A directed k -simplex is an ordered tuple $(v_0, \dots, v_k)$ with an edge $v_a \rightarrow v_b$ for all $a < b$. Edges exist only from later to earlier tokens, so a vertex set S supports exactly one directed simplex, its elements in decreasing order; every other ordering uses a missing forward edge. (A missing edge is not a zero-weight edge; encoding it as weight 0 would admit all orderings, a common implementation error.) This bijection preserves faces. Its filtration value is $\max_{a < b} (1 - A_{v_a v_b})$, and for $u > v$, $D_{uv} = 1 - A_{uv}$, so the undirected value $\max_{u, v \in S} D_{uv}$ is the same number. $\square$

We verified the bijection and the filtration identity by explicit enumeration on 420 random causal matrices ($N \leq 8$, four families including tie-dense and underflowing matrices; 30,060 vertex sets; zero violations), and confirmed that the identity fails for bidirectional attention, where a set of size n supports $n!$ orderings. The collapse is a consequence of causality: for decoders, the directed flag complex adds nothing to its symmetrized counterpart. Directional features that do not come from flag complexes, such as those of Kushnareva et al. (2021) for bidirectional encoders, are outside this statement.

B IMPLEMENTATION AND PROTOCOL DETAILS

Extraction. All reported settings run in bfloat16 on Apple-silicon (MPS) with eager attention; float16 on MPS produced non-finite values and degenerate generations for Qwen models, and the extractor refuses non-finite logits. Model identifiers are in Table 7 (Hugging Face main revisions as of September 2026). For each example we run one forward pass with attentions and hidden states, average heads, and form D in float64. Ripser (Bauer, 2021) computes 0D/1D barcodes; per-head MSTs use a batched dense Prim’s algorithm (SciPy’s sparse MST drops explicit zero-weight edges, which occur when $A_{u0} \approx 1$). δ_0 uses the closed form of Corollary 2; $\min_s \delta_s$ is computed by brute force over all apexes. DEFLATED deletes token 0 and renormalizes each row over the remaining causal support. ANSWER-ONLY restricts D to the answer tokens without renormalization. The Lookback ratio of head h is the mean over answer tokens t of $\bar{a}_{\text{ctx}}/(\bar{a}_{\text{ctx}} + \bar{a}_{\text{new}})$, with $\bar{a}$ the mean attention from t to prompt tokens and to answer tokens up to t . The LLM-Check score of a head is the mean log diagonal attention (the log-determinant of the triangular attention kernel divided by N), summed over heads per layer. Log-probability features are the mean, minimum, and sum of answer-token log-probabilities, mean and maximum predictive entropy, and mean maximum softmax probability. Hidden states are mean-pooled over answer tokens at layer $\lfloor L/2 \rfloor$.

Prompts. TruthfulQA uses each model’s own format: the Qwen chat template with an explicit system turn “You are a helpful assistant.”, `<|user|>...<|end|>` for Phi-3, `<|user|>...</s>` for TinyLlama, ChatML without system turn for SmoLLM, and `[INST]...[/INST]` for Mistral. HaluEval uses Knowledge: ... \nQuestion: ... \nAnswer: without a chat template. The prompt boundary is computed by tokenizing the prefix without its trailing space, which the full string merges into the first answer token. The on-policy prompt is the model’s chat template around “Answer the following question with a short phrase only, no explanation. Question: ...”.

Token 0 is whatever token the format places first (BOS for TinyLlama and Mistral, `<|user|>` for Phi-3, whose tokenizer adds no BOS, `<|im_start|>` for Qwen and SmolLM, and “Knowledge” for Qwen on HaluEval); the theorem does not require it to be a special token. On-policy answers were generated from the chat-formatted prompt with `add_special_tokens=False`, and features were extracted with the tokenizer’s default special tokens; the two differ only for TinyLlama, whose extraction input starts with a BOS token.

Sensitivity to inference precision. An earlier run of five settings in float16 lets us measure how much each bank depends on numerical precision (Table 4). Switching to bfloat16 changes the AUC of head-averaged topological banks by up to 0.058 (TQA Qw3B; at most 0.005 elsewhere), and of sink features by up to 0.003, but that of non-topological banks by at most 0.001. Short 1D bars live near the scale of δ_0 and are sensitive to small perturbations of attention, which is one more reason to report the precision alongside any attention-topology result.

Table 4: **AUC change from float16 to bfloat16** (same prompts and protocol; HaluEval Qwen2.5-3B also changed its prompt boundary, see above).

Setting	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw3B	−0.003	−0.018	+0.058	−0.001	−0.001	−0.001	0.000	0.000	−0.001
TQA Phi-3	−0.001	−0.001	−0.003	0.000	−0.001	0.000	0.000	0.000	0.000
TQA TinyLL	0.000	0.000	−0.001	−0.001	0.000	0.000	0.000	0.000	0.000
TQA SmolLM	0.000	−0.001	−0.001	+0.001	+0.002	0.000	0.000	0.000	0.000
HE Qw3B	+0.001	−0.005	+0.001	+0.002	−	−0.001	−	0.000	−

float32 against bfloat16. The fp16/bf16 comparison above varies precision at fixed hardware. To separate precision from platform we re-extracted 2 settings twice on one CUDA machine, in bfloat16 and in float32, and compare both with the bfloat16 numbers the paper reports, which were extracted on Apple-silicon MPS. The platform gap (bf16 CUDA against bf16 MPS, and so also torch and scikit-learn versions) is at most 0.005 AUC, and the precision gap (float32 against bfloat16 on the same machine) at most 0.007, 0.007 on the topological banks. Both are well inside the ± 0.015 equivalence margin, so the reported numbers reproduce on other hardware and float32 does not change a conclusion (Table 5).

Table 5: **Platform and precision sensitivity (Δ AUC).** “bf16 CUDA – bf16 MPS” isolates hardware and library versions at fixed precision; “fp32 – bf16 (CUDA)” isolates precision at fixed hardware.

Setting	Contrast	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw1.5B	bf16 CUDA – bf16 MPS	+0.001	+0.005	+0.003	+0.003	−0.002	0.000	+0.001	0.000	0.000
TQA Qw1.5B	fp32 – bf16 (CUDA)	+0.002	+0.004	+0.001	−0.007	+0.005	+0.003	+0.001	−0.001	0.000
TQA TinyLL	bf16 CUDA – bf16 MPS	0.000	−0.001	+0.001	−0.001	+0.001	0.000	0.000	0.000	0.000
TQA TinyLL	fp32 – bf16 (CUDA)	0.000	+0.001	+0.002	+0.003	−0.001	0.000	0.000	0.000	0.000

On-policy benchmark. Table 6 summarizes the generations each model produced and how they were graded.

Table 6: **TriviaQA on-policy benchmark.** 2,000 validation questions sampled with seed 0; greedy decoding, first line of the answer, at most 24 new tokens. An answer is correct if any normalized gold alias (lower-cased, punctuation and articles removed) occurs in the normalized answer as a whole-word span. Empty answers are dropped.

Model	questions	accuracy	rows used	mean answer tokens
Qwen2.5-3B	2,000	42.4%	2,000	5.5
Qwen2.5-1.5B	2,000	31.4%	2,000	5.8
Phi-3-mini	2,000	56.2%	2,000	6.0
TinyLlama-1.1B	2,000	26.4%	2,000	18.9
Mistral-7B	2,000	66.5%	2,000	13.0
Qwen2.5-7B	2,000	51.4%	2,000	5.3

Statistics. Bootstrap resamples are drawn over questions (both rows of a pair move together). TOST power at margin m assumes a true difference of 0 and a normal sampling distribution with the bootstrap SE: $2\Phi(m/\text{SE} - z_{0.95}) - 1$. Repeated-CV predictions (seeds 42, 43, 44) are averaged before bootstrapping (seed 0), so the intervals reflect sampling of questions for a fixed, split-averaged predictor and omit refitting variance. The seed-to-seed SD of the pooled AUC is reported separately (Appendix G); it is at most 0.008 and exceeds 0.004 in 22 of the bank-setting cells. The C rule depends on bank width, so the same bank can be fitted with $C=1$ for a shallow model and with tuned C for a deeper one. Late fusion fits the second-level model on first-level out-of-fold scores, as in standard stacking; its training rows’ first-level scores come from models that saw the corresponding test questions, which is a mild, standard optimism shared by both arms of the comparison.

C ADDITIONAL ANALYSES

Models. Table 7 lists the audited architectures. All models use grouped or multi-head attention; we average attention over query heads.

Table 7: Audited models.

Model	params	layers	heads	KV heads	hidden
Qwen/Qwen2.5-3B-Instruct	3.09B	36	16	2	2048
Qwen/Qwen2.5-1.5B-Instruct	1.54B	28	12	2	1536
microsoft/Phi-3-mini-4k-instruct	3.82B	32	32	32	3072
TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.10B	22	32	4	2048
HuggingFaceTB/SmolLM-1.7B-Instruct	1.71B	24	32	32	2048
mistralai/Mistral-7B-Instruct-v0.2	7.24B	32	32	8	4096

Length scaling of 1D persistence on real attention. Table 8 regresses log total 1D persistence (summed over layers) on log N with a class fixed effect, on rows with $P_1 > 0$, so that the exponent is not driven by a length difference between correct and hallucinated answers. Exponents range from 0.03 to 1.39, against 1.31 for i.i.d. uniform weights and 1.27 for random causal attention without a sink over the same range of N . 10 of 11 settings grow significantly more slowly than the i.i.d. reference (not HE Qw3B). The exponent exceeds 1 in HE Qw3B, HE Qw1.5B, so growth on real attention is not uniformly sub-linear. Summed over layers, almost every example has some 1D bar, because every model has some layers that are not coned; within an example, the coned layers contribute nothing to P_1 whatever the length (Table 1).

Table 8: Length scaling of 1D persistence. OLS of log P_1 on log N and the class label; rows with $P_1 = 0$ are excluded and counted in the first column.

Setting	rows with $P_1 > 0$	$\hat{\alpha}$ (class FE)	95% CI	R^2	Spearman(P_1, N)
TQA Qw3B	100.0%	0.25	[0.23, 0.27]	0.27	0.57
TQA Qw1.5B	100.0%	0.57	[0.52, 0.61]	0.29	0.51
TQA Phi-3	99.9%	0.93	[0.82, 1.04]	0.15	0.39
TQA TinyLL	100.0%	0.32	[0.28, 0.36]	0.14	0.28
TQA SmolLM	100.0%	0.61	[0.54, 0.69]	0.13	0.36
HE Qw3B	100.0%	1.39	[1.31, 1.47]	0.36	0.60
HE Qw1.5B	100.0%	1.10	[1.07, 1.13]	0.84	0.91
TrQA Qw3B	100.0%	0.72	[0.68, 0.75]	0.46	0.68
TrQA Qw1.5B	100.0%	0.75	[0.72, 0.78]	0.59	0.68
TrQA Phi-3	100.0%	0.70	[0.65, 0.75]	0.29	0.49
TrQA TinyLL	100.0%	0.03	[0.02, 0.03]	0.03	0.15
i.i.d. uniform weights (simulation)	100%	1.31	—	—	—
random causal attention, no sink (simulation)	100%	1.27	—	—	—

Sequence lengths. Table 9 gives class-conditional lengths. TruthfulQA is length-balanced by construction; HaluEval’s hallucinated answers are systematically longer; on-policy, incorrect answers differ in length from correct ones because they are the model’s own.

Table 9: **Class-conditional sequence length** (mean $\pm$ SD, full sequence) and mean answer length.

Setting	N (correct)	N (hallucinated)	answer tokens (c./h.)	point-biserial r	Welch p
TQA Qw3B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Qw1.5B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Phi-3	29.2 $\pm$ 8.9	28.7 $\pm$ 8.8	11.8 / 11.4	-0.024	0.32
TQA TinyLL	42.5 $\pm$ 9.0	42.1 $\pm$ 8.9	12.1 / 11.7	-0.024	0.32
TQA SmolLM	32.6 $\pm$ 8.2	32.1 $\pm$ 8.2	10.6 / 10.2	-0.026	0.29
HE Qw3B	121.2 $\pm$ 39.7	132.7 $\pm$ 41.2	3.8 / 15.4	+0.141	$< 10^{-9}$
HE Qw1.5B	122.2 $\pm$ 39.5	133.9 $\pm$ 41.3	3.6 / 15.3	+0.144	$< 10^{-5}$
TrQA Qw3B	68.0 $\pm$ 10.4	69.0 $\pm$ 9.7	5.3 / 5.7	+0.052	0.022
TrQA Qw1.5B	67.5 $\pm$ 10.3	69.4 $\pm$ 10.0	5.4 / 5.9	+0.086	0.00013
TrQA Phi-3	45.6 $\pm$ 11.3	46.9 $\pm$ 11.2	5.6 / 6.5	+0.057	0.01
TrQA TinyLL	67.3 $\pm$ 12.5	73.7 $\pm$ 14.9	17.0 / 19.6	+0.196	$< 10^{-20}$

Margin sensitivity. Table 10 reports TOST p -values at four equivalence margins for the four comparisons of Figure 5; bold marks $p < 0.05$ (equivalence).

Table 10: **TOST p -values by margin m .** NT: all non-topological features.

Setting m	Sink vs. 0D				0D Sink				1D 0D				0D NT			
	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025
TQA Qw3B	.029	.001	<.001	<.001	.040	.001	<.001	<.001	1.000	.995	.972	.882	<.001	<.001	<.001	<.001
TQA Qw1.5B	.109	.016	<.001	<.001	.692	.276	.040	.002	.311	.034	<.001	<.001	<.001	<.001	<.001	<.001
TQA Phi-3	.138	.013	<.001	<.001	.112	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA TinyLL	.520	.234	.064	.012	.962	.809	.483	.169	.210	.021	<.001	<.001	<.001	<.001	<.001	<.001
TQA SmolLM	.939	.865	.748	.602	1.000	.996	.960	.805	.179	.036	.003	<.001	<.001	<.001	<.001	<.001
HE Qw3B	.910	.676	.351	.110	1.000	1.000	.982	.804	.906	.551	.148	.012	<.001	<.001	<.001	<.001
HE Qw1.5B	.993	.970	.896	.755	.679	.263	.041	.002	.980	.918	.786	.569	<.001	<.001	<.001	<.001
TrQA Qw3B	.792	.522	.239	.069	.982	.866	.561	.219	.106	.015	.002	<.001	<.001	<.001	<.001	<.001
TrQA Qw1.5B	.431	.176	.050	.010	.528	.211	.044	.005	.079	.011	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Phi-3	.047	.002	<.001	<.001	.114	.005	<.001	<.001	.086	.003	<.001	<.001	<.001	<.001	<.001	<.001
TrQA TinyLL	.091	.021	.003	<.001	.071	.005	<.001	<.001	.083	.013	.002	<.001	<.001	<.001	<.001	<.001

Per-layer signal. Figure 2 shows single-feature AUCs by depth. P_0 and the star weight trace nearly identical curves in every model with a sink, while P_1 carries signal only in the layers that are not coned. These AUCs are orientation-free ($\max(\text{AUC}, 1 - \text{AUC})$) and therefore slightly optimistic; they are descriptive, not a detector.

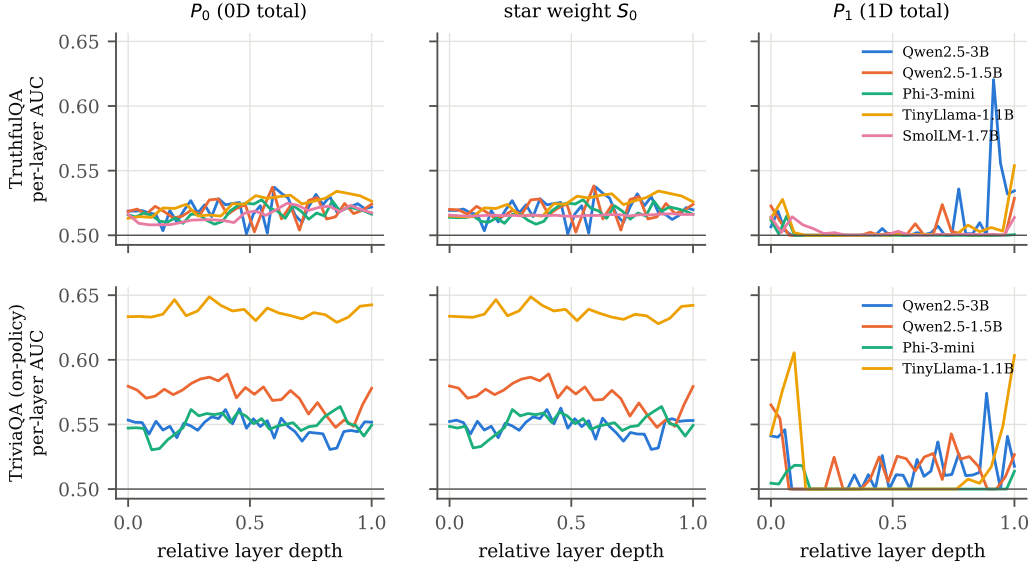


Figure 2: Single-feature AUC by layer.

Worked example. Figure 3 shows TruthfulQA question 67 with its incorrect answer (“Yes, vampires are real”) under TinyLlama-1.1B ($N=29$). In layer 5 the model sends 0.85 of its attention to token 0 on average, $\delta_0 = 0.000$, and accordingly $P_0 = 4.225$ equals the star weight $S_0 = 4.225$ and there are 0 H_1 bars. In layer 21 sink attention drops to 0.53 and $\delta_0 = 0.281$; now $P_0 = 12.287$ falls below $S_0 = 13.043$ (within the bound $(N-1)\delta_0 = 7.87$), and 10 H_1 bars appear, the longest of length $0.172 \leq \delta_0$. The theorem bounds loops but does not force them: in layers 2 and 11–18 of the same example δ_0 lies between 0.11 and 0.52, yet H_1 is empty.

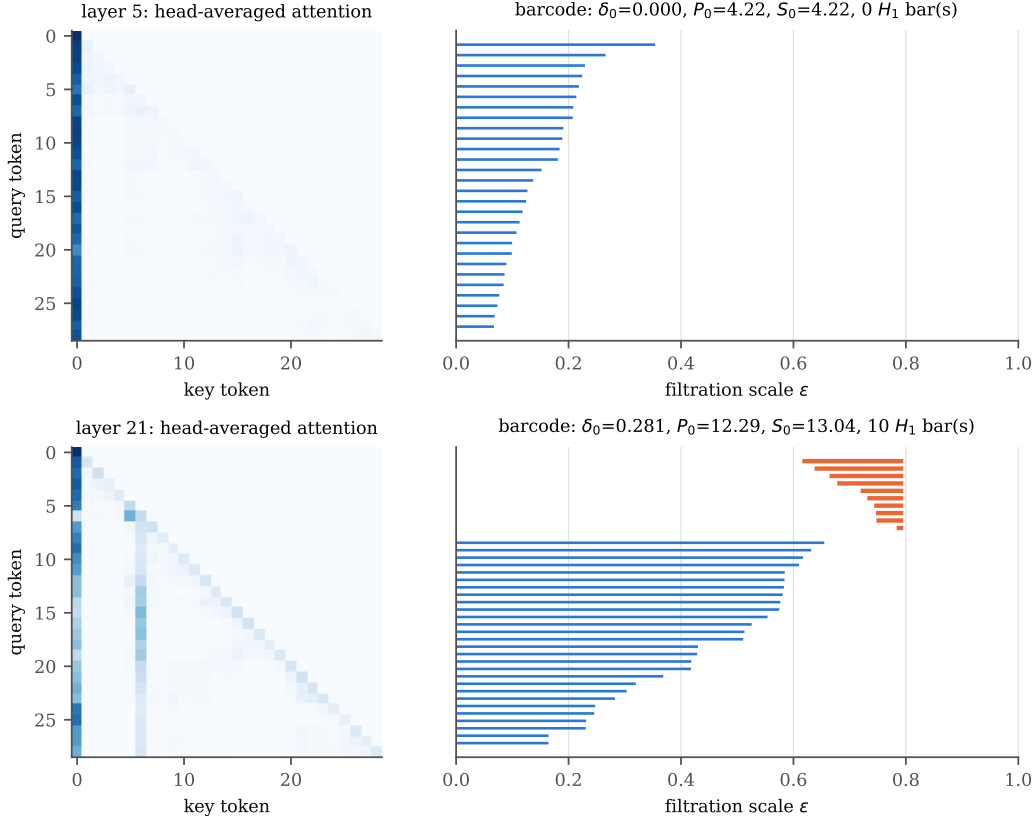


Figure 3: **A coned and a non-coned layer of the same example** (TinyLlama-1.1B, TruthfulQA). Left: head-averaged attention; the bright first column is the sink. Right: 0D bars (blue) and 1D bars (orange) of the Vietoris-Rips filtration of $D = 1 - \max(A, A^T)$.

D ONE-DIMENSIONAL PERSISTENCE

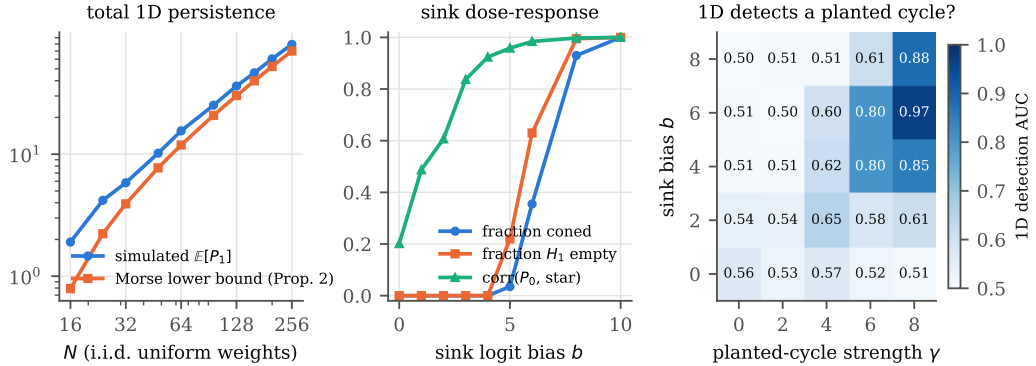


Figure 4: **Synthetic checks.** Left: Proposition 2’s lower bound against the simulated mean total 1D persistence under i.i.d. Uniform $[0, 1]$ weights (20 draws per $N \leq 128$, 8 above). Middle: random causal attention ($N=40$, 200 draws per point) with a logit bias b on token 0: as the sink strengthens, graphs become exactly coned, H_1 empties, and P_0 becomes a function of the star weight. Right: AUC of 1D total persistence for detecting a planted 6-cycle of logit strength γ (150 draws per class). Without a sink ($b=0$) the planted cycle is lost among background cycles; under a strong sink ($b=8$) it is invisible unless it is strong enough to break the cone.

Under i.i.d. weights, 1D persistence grows faster than linearly over the lengths that matter here and linearly asymptotically (Proposition 2; Figure 4, left). Real attention behaves very differently: most graphs have *no* 1D bars at all (Table 1), and the bars that exist are confined to graphs with a non-zero coning defect. Figure 4 (middle) reproduces this transition in a controlled setting: raising the sink’s logit bias moves random causal attention from the i.i.d.-like regime to exact coning, with H_1 emptying in lockstep.

This explains a recurring empirical observation, that 1D features add little to 0D. At full sample size, adding length-normalized 1D to 0D changes AUC by -0.008 to $+0.033$ and is equivalent within ± 0.015 in 9 of 13 sink settings; the exceptions are TQA Qw3B, HE Qw3B, HE Qw1.5B and TrQA Mis7B, where adding 1D helps significantly. The TruthfulQA Qwen2.5-3B gain is also the result that is most sensitive to inference precision (Appendix B). The mechanism is not that loops are uninformative in principle. In a strongly sinked model, a loop can only exist where some token pays another token more attention than one of the two pays to the sink, and its lifetime is bounded by that excess. Figure 4 (right) makes this concrete. A planted 6-cycle that H_1 detects easily under a moderate sink ($b=6$, $\gamma=8$: AUC 0.97) is invisible under a strong one ($b=8$, $\gamma \leq 4$: AUC ≤ 0.51) until it is strong enough to break the cone. Without any sink ($b=0$) H_1 also misses it (AUC ≤ 0.57), because random attention already has many background cycles; 1D persistence is informative only in between.

E RECOMMENDATIONS FOR REPORTING ATTENTION-GRAPH TOPOLOGY

(i) Report sink attention and δ_0 (or the relevant defect, such as δ_P for TOHA) next to any attention-graph topological feature, and compare the detector with the first-order statistic that the reduction names. (ii) Compare topological detectors against per-head non-topological statistics (sink mass, entropy, Lookback ratios) at the same resolution, and against a hidden-state probe. (iii) Evaluate on the model’s own generations; teacher-forced benchmarks mix detection with recognition of dataset authorship. (iv) If structure beyond the sink is the goal, remove the sink first (deflation) or restrict to answer tokens. Both expose real non-sink geometry, but in our settings it was not new information.

F VALIDATION OF THE TOHA IMPLEMENTATION AND THE ON-POLICY LABELS

Against the authors’ released code. Our per-head score d is our own implementation, so we check it against the routine the TOHA authors release (Bazarova et al., 2026; `src/methods/mtopdiv/utis.py`), which builds the same distance matrix and calls `ripser` rather than our dense Prim’s algorithm. Over 21,696 head graphs from 2 settings the two agree to 1.2×10^{-8} (Table 11), so the reduction in Proposition 1 is not an artefact of how we compute d .

Table 11: **Our d against the authors’ MTop-Div code.** Δ is the per-head difference; both are divided by $|R|$.

Setting	head graphs	max $ \Delta $	mean $ \Delta $	Pearson r
TQA Mis7B	12,288	0	0	1.00000000
TQA Qw7B	9,408	1.2×10^{-8}	1.1×10^{-11}	1.00000000

LLM-judge audit of the on-policy labels. The on-policy label is an alias string match, which mislabels paraphrases and hedged answers. We sampled 200 answers per setting (seed 0) in 2 settings and had Qwen2.5-7B-Instruct grade them from the question, the gold aliases and the answer. Disagreement is 6% to 14% (Table 12), and is mostly the string match being too lax. Re-scoring the same out-of-fold predictions on those rows under the judge’s labels shifts every bank by at most 0.058 AUC and leaves the dominance orderings intact, so the conclusions do not rest on the labelling. The judge is itself a language model, not a human annotator, and one of the audited settings is graded by the same model that produced its answers, so 14% should be read as a floor rather than an estimate.

Table 12: **LLM-judge audit of the on-policy labels.** “match 0/judge 1”: the string match calls it hallucinated and the judge does not. Last column: largest change in any bank’s AUC on the audited rows when the judge’s labels replace the string match.

Setting	judged	disagree	match 0/judge 1	match 1/judge 0	max $ \Delta\text{AUC} $
TrQA Mis7B	200	14.0%	8	20	0.058
TrQA Qw7B	200	5.5%	7	4	0.027

G FULL RESULTS

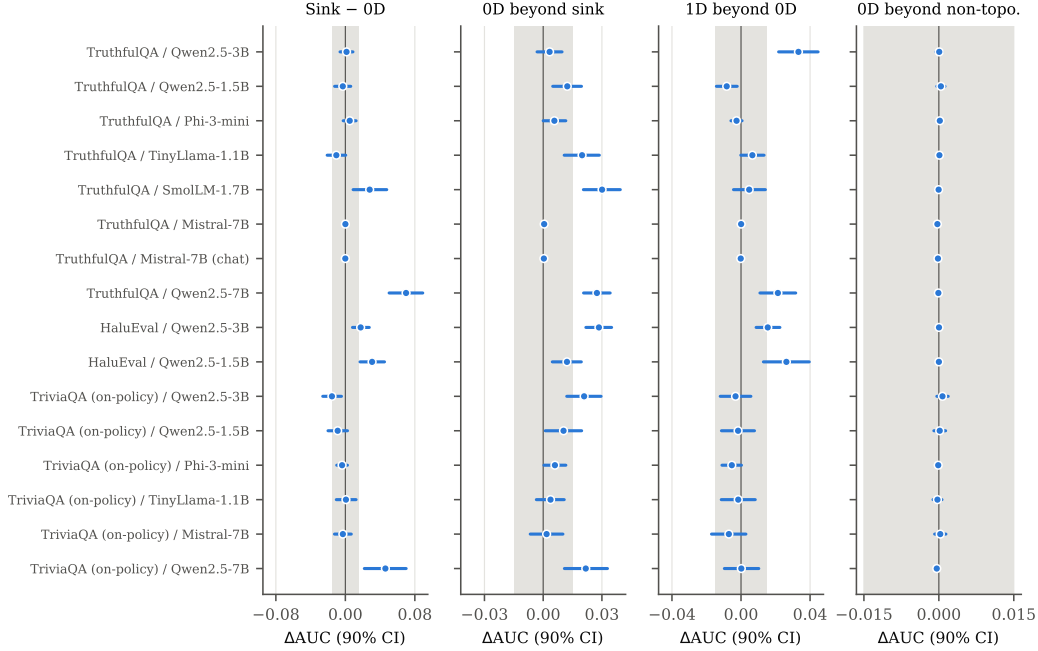


Figure 5: **Four comparisons across all settings** (ΔAUC with 90% cluster-bootstrap intervals; shaded band: ± 0.015). Sink features nearly reproduce 0D; 0D adds little beyond them; 1D adds little beyond 0D except in three settings; and 0D adds nothing beyond non-topological probes under early fusion (late fusion in Table 13).

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	LEN	2	0.529	0.0003	0.567
TQA Qw3B	LEX	—	0.820	0.0014	0.829
TQA Qw3B	LOGPROB	6	0.630	0.0006	0.694
TQA Qw3B	LLMCHECK	36	0.693	0.0011	0.764
TQA Qw3B	ROWSTAT	72	0.750	0.0003	0.823
TQA Qw3B	SINK	72	0.737	0.0022	0.767
TQA Qw3B	0D	72	0.736	0.0013	0.778
TQA Qw3B	LEGACY MSTPROXY	36	0.713	0.0008	0.788
TQA Qw3B	1D	144	0.660	0.0001	0.674
TQA Qw3B	DEFL	216	0.715	0.0028	0.753
TQA Qw3B	ANS	216	0.748	0.0017	0.749
TQA Qw3B	PH SINK	576	0.851	0.0015	0.870
TQA Qw3B	PH 0DTOT	576	0.842	0.0011	0.869
TQA Qw3B	PH 0D	1152	0.853	0.0013	0.862
TQA Qw3B	PH ENT	576	0.857	0.0013	0.901
TQA Qw3B	LOOKBACK	576	0.897	0.0022	0.908
TQA Qw3B	HIDDEN	2048	0.913	0.0008	0.918
TQA Qw3B	NONTOPO	2738	0.926	0.0006	0.936
TQA Qw3B	0D+1D	216	0.769	0.0004	0.791

	Setting	Bank	dim	AUC	seed SD	pair acc.
1242						
1243						
1244	TQA Qw3B	0D+SINK	144	0.741	0.0015	0.776
1245	TQA Qw3B	HIDDEN+0D	2120	0.913	0.0007	0.920
1246	TQA Qw3B	LEX+0D	–	0.840	0.0015	0.848
1247	TQA Qw3B	LEX+DEFL	–	0.830	0.0004	0.854
1248	TQA Qw3B	LEX+SINK	–	0.843	0.0020	0.851
1249	TQA Qw3B	LOGPROB+0D	78	0.776	0.0022	0.810
1250	TQA Qw3B	NONTOPO+0D	2810	0.926	0.0004	0.936
1251	TQA Qw3B	NONTOPO+DEFL	2954	0.928	0.0006	0.933
1252	TQA Qw3B	PH SINK+PH 0DTOT	1152	0.860	0.0022	0.892
1253	TQA Qw3B	SINK+1D	216	0.766	0.0007	0.787
1254	TQA Qw3B	SINK+ANS	288	0.785	0.0031	0.797
1255	TQA Qw3B	SINK+DEFL	288	0.766	0.0024	0.799
1256	TQA Qw1.5B	LEN	2	0.529	0.0003	0.567
1257	TQA Qw1.5B	LEX	–	0.820	0.0014	0.829
1258	TQA Qw1.5B	LOGPROB	6	0.638	0.0007	0.682
1259	TQA Qw1.5B	LLMCHECK	28	0.666	0.0005	0.743
1260	TQA Qw1.5B	ROWSTAT	56	0.738	0.0014	0.815
1261	TQA Qw1.5B	SINK	56	0.723	0.0019	0.771
1262	TQA Qw1.5B	0D	56	0.726	0.0007	0.767
1263	TQA Qw1.5B	LEGACY MSTPROXY	28	0.698	0.0010	0.786
1264	TQA Qw1.5B	1D	112	0.586	0.0014	0.644
1265	TQA Qw1.5B	DEFL	168	0.703	0.0042	0.755
1266	TQA Qw1.5B	ANS	168	0.697	0.0040	0.700
1267	TQA Qw1.5B	PH SINK	336	0.830	0.0008	0.851
1268	TQA Qw1.5B	PH 0DTOT	336	0.819	0.0011	0.838
1269	TQA Qw1.5B	PH 0D	672	0.838	0.0012	0.847
1270	TQA Qw1.5B	PH ENT	336	0.833	0.0025	0.852
1271	TQA Qw1.5B	LOOKBACK	336	0.851	0.0020	0.856
1272	TQA Qw1.5B	HIDDEN	1536	0.916	0.0012	0.923
1273	TQA Qw1.5B	NONTOPO	1962	0.917	0.0013	0.928
1274	TQA Qw1.5B	0D+1D	168	0.718	0.0013	0.767
1275	TQA Qw1.5B	0D+SINK	112	0.735	0.0012	0.766
1276	TQA Qw1.5B	HIDDEN+0D	1592	0.916	0.0014	0.924
1277	TQA Qw1.5B	LEX+0D	–	0.835	0.0017	0.841
1278	TQA Qw1.5B	LEX+DEFL	–	0.825	0.0019	0.836
1279	TQA Qw1.5B	LEX+SINK	–	0.833	0.0008	0.846
1280	TQA Qw1.5B	LOGPROB+0D	62	0.740	0.0013	0.797
1281	TQA Qw1.5B	NONTOPO+0D	2018	0.917	0.0014	0.927
1282	TQA Qw1.5B	NONTOPO+DEFL	2130	0.917	0.0014	0.923
1283	TQA Qw1.5B	PH SINK+PH 0DTOT	672	0.841	0.0013	0.867
1284	TQA Qw1.5B	SINK+1D	168	0.715	0.0017	0.769
1285	TQA Qw1.5B	SINK+ANS	224	0.761	0.0027	0.753
1286	TQA Qw1.5B	SINK+DEFL	224	0.757	0.0028	0.802
1287	TQA Phi-3	LEN	2	0.522	0.0004	0.562
1288	TQA Phi-3	LEX	–	0.820	0.0014	0.829
1289	TQA Phi-3	LOGPROB	6	0.578	0.0016	0.597
1290	TQA Phi-3	LLMCHECK	32	0.653	0.0008	0.755
1291	TQA Phi-3	ROWSTAT	64	0.661	0.0011	0.738
1292	TQA Phi-3	SINK	64	0.651	0.0006	0.700
1293	TQA Phi-3	0D	64	0.646	0.0011	0.709
1294	TQA Phi-3	LEGACY MSTPROXY	32	0.613	0.0008	0.694
1295	TQA Phi-3	1D	128	0.526	0.0030	0.551
	TQA Phi-3	DEFL	192	0.639	0.0011	0.684
	TQA Phi-3	ANS	192	0.629	0.0042	0.651
	TQA Phi-3	PH SINK	1024	0.871	0.0008	0.901
	TQA Phi-3	PH 0DTOT	1024	0.845	0.0014	0.882
	TQA Phi-3	PH 0D	2048	0.868	0.0007	0.881
	TQA Phi-3	PH ENT	1024	0.864	0.0010	0.895
	TQA Phi-3	LOOKBACK	1024	0.892	0.0004	0.902
	TQA Phi-3	HIDDEN	3072	0.933	0.0007	0.945
	TQA Phi-3	NONTOPO	4198	0.936	0.0006	0.942
	TQA Phi-3	0D+1D	192	0.643	0.0014	0.705
	TQA Phi-3	0D+SINK	128	0.657	0.0007	0.704

	Setting	Bank	dim	AUC	seed SD	pair acc.
1296						
1297						
1298	TQA Phi-3	HIDDEN+0D	3136	0.933	0.0008	0.944
1299	TQA Phi-3	LEX+0D	–	0.821	0.0003	0.842
1299	TQA Phi-3	LEX+DEFL	–	0.802	0.0001	0.827
1300	TQA Phi-3	LEX+SINK	–	0.820	0.0007	0.834
1301	TQA Phi-3	LOGPROB+0D	70	0.655	0.0011	0.694
1302	TQA Phi-3	NONTOPO+0D	4262	0.936	0.0006	0.942
1303	TQA Phi-3	NONTOPO+DEFL	4390	0.937	0.0007	0.939
1304	TQA Phi-3	PH SINK+PH 0DTOT	2048	0.871	0.0007	0.905
1304	TQA Phi-3	SINK+1D	192	0.651	0.0011	0.696
1305	TQA Phi-3	SINK+ANS	256	0.675	0.0015	0.698
1306	TQA Phi-3	SINK+DEFL	256	0.684	0.0023	0.718
1307	TQA TinyLl.	LEN	2	0.523	0.0005	0.556
1308	TQA TinyLl.	LEX	–	0.820	0.0014	0.829
1308	TQA TinyLl.	LOGPROB	6	0.637	0.0010	0.645
1309	TQA TinyLl.	LLMCHECK	22	0.614	0.0005	0.705
1310	TQA TinyLl.	ROWSTAT	44	0.661	0.0015	0.749
1311	TQA TinyLl.	SINK	44	0.643	0.0006	0.718
1312	TQA TinyLl.	0D	44	0.654	0.0011	0.729
1313	TQA TinyLl.	LEGACY MSTPROXY	22	0.627	0.0009	0.741
1314	TQA TinyLl.	1D	88	0.592	0.0006	0.673
1315	TQA TinyLl.	DEFL	132	0.651	0.0019	0.705
1315	TQA TinyLl.	ANS	132	0.644	0.0049	0.678
1316	TQA TinyLl.	PH SINK	704	0.808	0.0015	0.841
1317	TQA TinyLl.	PH 0DTOT	704	0.804	0.0021	0.830
1318	TQA TinyLl.	PH 0D	1408	0.806	0.0015	0.820
1319	TQA TinyLl.	PH ENT	704	0.817	0.0008	0.838
1319	TQA TinyLl.	LOOKBACK	704	0.850	0.0028	0.846
1320	TQA TinyLl.	HIDDEN	2048	0.883	0.0013	0.879
1321	TQA TinyLl.	NONTOPO	2824	0.885	0.0011	0.887
1322	TQA TinyLl.	0D+1D	132	0.660	0.0010	0.744
1323	TQA TinyLl.	0D+SINK	88	0.663	0.0004	0.728
1324	TQA TinyLl.	HIDDEN+0D	2092	0.883	0.0014	0.880
1324	TQA TinyLl.	LEX+0D	–	0.830	0.0012	0.836
1325	TQA TinyLl.	LEX+DEFL	–	0.826	0.0012	0.846
1326	TQA TinyLl.	LEX+SINK	–	0.831	0.0011	0.832
1327	TQA TinyLl.	LOGPROB+0D	50	0.683	0.0010	0.711
1328	TQA TinyLl.	NONTOPO+0D	2868	0.885	0.0011	0.889
1328	TQA TinyLl.	NONTOPO+DEFL	2956	0.884	0.0010	0.885
1329	TQA TinyLl.	PH SINK+PH 0DTOT	1408	0.810	0.0031	0.841
1330	TQA TinyLl.	SINK+1D	132	0.651	0.0016	0.729
1331	TQA TinyLl.	SINK+ANS	176	0.685	0.0041	0.720
1332	TQA TinyLl.	SINK+DEFL	176	0.688	0.0015	0.749
1333	TQA SmolLM	LEN	2	0.524	0.0003	0.554
1334	TQA SmolLM	LEX	–	0.820	0.0014	0.829
1334	TQA SmolLM	LOGPROB	6	0.588	0.0010	0.592
1335	TQA SmolLM	LLMCHECK	24	0.604	0.0004	0.684
1336	TQA SmolLM	ROWSTAT	48	0.673	0.0014	0.741
1337	TQA SmolLM	SINK	48	0.611	0.0019	0.628
1338	TQA SmolLM	0D	48	0.583	0.0029	0.656
1339	TQA SmolLM	LEGACY MSTPROXY	24	0.585	0.0011	0.674
1339	TQA SmolLM	1D	96	0.543	0.0015	0.581
1340	TQA SmolLM	DEFL	144	0.594	0.0023	0.650
1341	TQA SmolLM	ANS	144	0.628	0.0041	0.630
1342	TQA SmolLM	PH SINK	768	0.720	0.0008	0.775
1343	TQA SmolLM	PH 0DTOT	768	0.819	0.0019	0.845
1344	TQA SmolLM	PH 0D	1536	0.800	0.0012	0.818
1344	TQA SmolLM	PH ENT	768	0.834	0.0007	0.841
1345	TQA SmolLM	LOOKBACK	768	0.863	0.0017	0.864
1346	TQA SmolLM	HIDDEN	2048	0.894	0.0022	0.892
1347	TQA SmolLM	NONTOPO	2894	0.899	0.0015	0.903
1348	TQA SmolLM	0D+1D	144	0.588	0.0019	0.643
1348	TQA SmolLM	0D+SINK	96	0.641	0.0026	0.662
1349	TQA SmolLM	HIDDEN+0D	2096	0.894	0.0020	0.891

	Setting	Bank	dim	AUC	seed SD	pair acc.
1350						
1351						
1352	TQA SmolLM	LEX+0D	–	0.827	0.0008	0.838
1353	TQA SmolLM	LEX+DEFL	–	0.822	0.0001	0.845
1354	TQA SmolLM	LEX+SINK	–	0.819	0.0015	0.826
1355	TQA SmolLM	LOGPROB+0D	54	0.613	0.0009	0.630
1356	TQA SmolLM	NONTOPO+0D	2942	0.899	0.0013	0.902
1357	TQA SmolLM	NONTOPO+DEFL	3038	0.899	0.0012	0.900
1358	TQA SmolLM	PH SINK+PH 0DTOT	1536	0.826	0.0023	0.854
1359	TQA SmolLM	SINK+1D	144	0.629	0.0013	0.650
1360	TQA SmolLM	SINK+ANS	192	0.646	0.0021	0.651
1361	TQA SmolLM	SINK+DEFL	192	0.652	0.0022	0.681
1362	TQA Mis7B	LEN	2	0.525	0.0002	0.571
1363	TQA Mis7B	LEX	–	0.820	0.0008	0.828
1364	TQA Mis7B	LOGPROB	6	0.641	0.0004	0.652
1365	TQA Mis7B	LLMCHECK	32	0.652	0.0002	0.770
1366	TQA Mis7B	ROWSTAT	64	0.694	0.0009	0.821
1367	TQA Mis7B	SINK	64	0.685	0.0008	0.755
1368	TQA Mis7B	0D	64	0.684	0.0008	0.755
1369	TQA Mis7B	LEGACY MSTPROXY	32	0.646	0.0009	0.769
1370	TQA Mis7B	1D	128	0.501	0.0000	0.499
1371	TQA Mis7B	DEFL	192	0.710	0.0015	0.756
1372	TQA Mis7B	ANS	192	0.677	0.0043	0.704
1373	TQA Mis7B	PH SINK	1024	0.903	0.0009	0.938
1374	TQA Mis7B	PH 0DTOT	1024	0.887	0.0020	0.928
1375	TQA Mis7B	PH 0D	2048	0.893	0.0013	0.908
1376	TQA Mis7B	PH ENT	1024	0.897	0.0010	0.929
1377	TQA Mis7B	LOOKBACK	1024	0.933	0.0008	0.945
1378	TQA Mis7B	HIDDEN	4096	0.956	0.0005	0.957
1379	TQA Mis7B	NONTOPO	5222	0.959	0.0006	0.967
1380	TQA Mis7B	0D+1D	192	0.685	0.0007	0.760
1381	TQA Mis7B	0D+SINK	128	0.685	0.0007	0.754
1382	TQA Mis7B	HIDDEN+0D	4160	0.956	0.0005	0.956
1383	TQA Mis7B	LEX+0D	–	0.834	0.0013	0.859
1384	TQA Mis7B	LEX+DEFL	–	0.823	0.0013	0.856
1385	TQA Mis7B	LEX+SINK	–	0.833	0.0015	0.859
1386	TQA Mis7B	LOGPROB+0D	70	0.722	0.0012	0.775
1387	TQA Mis7B	NONTOPO+0D	5286	0.958	0.0006	0.967
1388	TQA Mis7B	NONTOPO+DEFL	5414	0.959	0.0006	0.966
1389	TQA Mis7B	PH SINK+PH 0DTOT	2048	0.903	0.0013	0.936
1390	TQA Mis7B	SINK+1D	192	0.685	0.0008	0.758
1391	TQA Mis7B	SINK+ANS	256	0.727	0.0023	0.763
1392	TQA Mis7B	SINK+DEFL	256	0.740	0.0004	0.800
1393	TQA Mis7B (chat)	LEN	2	0.525	0.0002	0.571
1394	TQA Mis7B (chat)	LEX	–	0.820	0.0008	0.828
1395	TQA Mis7B (chat)	LOGPROB	6	0.643	0.0010	0.646
1396	TQA Mis7B (chat)	LLMCHECK	32	0.653	0.0001	0.769
1397	TQA Mis7B (chat)	ROWSTAT	64	0.700	0.0008	0.823
1398	TQA Mis7B (chat)	SINK	64	0.688	0.0010	0.754
1399	TQA Mis7B (chat)	0D	64	0.688	0.0010	0.755
1400	TQA Mis7B (chat)	LEGACY MSTPROXY	32	0.652	0.0009	0.783
1401	TQA Mis7B (chat)	1D	128	0.501	0.0000	0.499
1402	TQA Mis7B (chat)	DEFL	192	0.733	0.0013	0.781
1403	TQA Mis7B (chat)	ANS	192	0.723	0.0033	0.736
	TQA Mis7B (chat)	PH SINK	1024	0.907	0.0011	0.934
	TQA Mis7B (chat)	PH 0DTOT	1024	0.890	0.0021	0.931
	TQA Mis7B (chat)	PH 0D	2048	0.900	0.0017	0.919
	TQA Mis7B (chat)	PH ENT	1024	0.901	0.0010	0.928
	TQA Mis7B (chat)	LOOKBACK	1024	0.930	0.0008	0.942
	TQA Mis7B (chat)	HIDDEN	4096	0.958	0.0004	0.956
	TQA Mis7B (chat)	NONTOPO	5222	0.960	0.0003	0.957
	TQA Mis7B (chat)	0D+1D	192	0.688	0.0009	0.754
	TQA Mis7B (chat)	0D+SINK	128	0.689	0.0008	0.755
	TQA Mis7B (chat)	HIDDEN+0D	4160	0.959	0.0004	0.957
	TQA Mis7B (chat)	LEX+0D	–	0.834	0.0013	0.860

	Setting	Bank	dim	AUC	seed SD	pair acc.
1404						
1405						
1406	TQA Mis7B (chat)	LEX+DEFL	–	0.835	0.0020	0.864
1407	TQA Mis7B (chat)	LEX+SINK	–	0.834	0.0015	0.859
1408	TQA Mis7B (chat)	LOGPROB+0D	70	0.729	0.0021	0.793
1409	TQA Mis7B (chat)	NONTOPO+0D	5286	0.960	0.0003	0.958
1410	TQA Mis7B (chat)	NONTOPO+DEFL	5414	0.961	0.0005	0.960
1411	TQA Mis7B (chat)	PH SINK+PH 0DTOT	2048	0.907	0.0012	0.938
1412	TQA Mis7B (chat)	SINK+1D	192	0.688	0.0009	0.756
1413	TQA Mis7B (chat)	SINK+ANS	256	0.751	0.0035	0.788
1414	TQA Mis7B (chat)	SINK+DEFL	256	0.756	0.0018	0.803
1415	TQA Qw7B	LEN	2	0.529	0.0003	0.567
1416	TQA Qw7B	LEX	–	0.820	0.0008	0.828
1417	TQA Qw7B	LOGPROB	6	0.674	0.0007	0.717
1418	TQA Qw7B	LLMCHECK	28	0.667	0.0011	0.729
1419	TQA Qw7B	ROWSTAT	56	0.736	0.0008	0.807
1420	TQA Qw7B	SINK	56	0.749	0.0001	0.761
1421	TQA Qw7B	0D	56	0.679	0.0010	0.774
1422	TQA Qw7B	LEGACY MSTPROXY	28	0.678	0.0011	0.785
1423	TQA Qw7B	1D	112	0.610	0.0005	0.634
1424	TQA Qw7B	DEFL	168	0.697	0.0009	0.748
1425	TQA Qw7B	ANS	168	0.741	0.0019	0.749
1426	TQA Qw7B	PH SINK	784	0.821	0.0025	0.843
1427	TQA Qw7B	PH 0DTOT	784	0.872	0.0016	0.909
1428	TQA Qw7B	PH 0D	1568	0.850	0.0022	0.867
1429	TQA Qw7B	PH ENT	784	0.887	0.0003	0.901
1430	TQA Qw7B	LOOKBACK	784	0.922	0.0004	0.923
1431	TQA Qw7B	HIDDEN	3584	0.933	0.0008	0.934
1432	TQA Qw7B	NONTOPO	4458	0.950	0.0013	0.950
1433	TQA Qw7B	0D+1D	168	0.701	0.0002	0.756
1434	TQA Qw7B	0D+SINK	112	0.777	0.0004	0.794
1435	TQA Qw7B	HIDDEN+0D	3640	0.933	0.0008	0.933
1436	TQA Qw7B	LEX+0D	–	0.834	0.0008	0.858
1437	TQA Qw7B	LEX+DEFL	–	0.828	0.0014	0.844
1438	TQA Qw7B	LEX+SINK	–	0.848	0.0001	0.849
1439	TQA Qw7B	LOGPROB+0D	62	0.727	0.0002	0.792
1440	TQA Qw7B	NONTOPO+0D	4514	0.949	0.0013	0.949
1441	TQA Qw7B	NONTOPO+DEFL	4626	0.949	0.0012	0.952
1442	TQA Qw7B	PH SINK+PH 0DTOT	1568	0.881	0.0005	0.920
1443	TQA Qw7B	SINK+1D	168	0.767	0.0023	0.780
1444	TQA Qw7B	SINK+ANS	224	0.794	0.0010	0.802
1445	TQA Qw7B	SINK+DEFL	224	0.788	0.0006	0.797
1446	HE Qw3B	LEN	2	0.962	0.0001	0.970
1447	HE Qw3B	LEX	–	0.981	0.0004	0.977
1448	HE Qw3B	LOGPROB	6	0.993	0.0001	0.991
1449	HE Qw3B	LLMCHECK	36	0.745	0.0004	0.956
1450	HE Qw3B	ROWSTAT	72	0.829	0.0002	0.976
1451	HE Qw3B	SINK	72	0.829	0.0008	0.930
1452	HE Qw3B	0D	72	0.811	0.0006	0.927
1453	HE Qw3B	LEGACY MSTPROXY	36	0.731	0.0002	0.960
1454	HE Qw3B	1D	144	0.710	0.0013	0.891
1455	HE Qw3B	DEFL	216	0.834	0.0009	0.959
1456	HE Qw3B	ANS	216	0.978	0.0009	0.980
1457	HE Qw3B	PH SINK	576	0.958	0.0007	0.999
	HE Qw3B	PH 0DTOT	576	0.958	0.0005	0.995
	HE Qw3B	PH 0D	1152	0.971	0.0006	0.988
	HE Qw3B	PH ENT	576	0.964	0.0005	0.993
	HE Qw3B	LOOKBACK	576	0.998	0.0001	0.998
	HE Qw3B	HIDDEN	2048	1.000	0.0000	1.000
	HE Qw3B	NONTOPO	2738	1.000	0.0002	1.000
	HE Qw3B	0D+1D	216	0.826	0.0013	0.945
	HE Qw3B	0D+SINK	144	0.857	0.0010	0.948
	HE Qw3B	HIDDEN+0D	2120	1.000	0.0000	1.000
	HE Qw3B	LEX+0D	–	0.979	0.0001	0.983
	HE Qw3B	LEX+DEFL	–	0.971	0.0003	0.975

	Setting	Bank	dim	AUC	seed SD	pair acc.
1458						
1459						
1460	HE Qw3B	LEX+SINK	–	0.980	0.0003	0.981
1461	HE Qw3B	LOGPROB+0D	78	0.993	0.0003	0.992
1462	HE Qw3B	NONTOPO+0D	2810	1.000	0.0002	1.000
1463	HE Qw3B	NONTOPO+DEFL	2954	1.000	0.0002	1.000
1464	HE Qw3B	PH SINK+PH 0DTOT	1152	0.962	0.0006	0.997
1465	HE Qw3B	SINK+1D	216	0.843	0.0011	0.938
1466	HE Qw3B	SINK+ANS	288	0.978	0.0003	0.982
1467	HE Qw3B	SINK+DEFL	288	0.887	0.0020	0.957
1468	HE Qw1.5B	LEN	2	0.967	0.0002	0.976
1469	HE Qw1.5B	LEX	–	0.979	0.0004	0.980
1470	HE Qw1.5B	LOGPROB	6	0.995	0.0002	0.988
1471	HE Qw1.5B	LLMCHECK	28	0.791	0.0005	0.978
1472	HE Qw1.5B	ROWSTAT	56	0.788	0.0017	0.984
1473	HE Qw1.5B	SINK	56	0.820	0.0015	0.934
1474	HE Qw1.5B	0D	56	0.789	0.0014	0.918
1475	HE Qw1.5B	LEGACY MSTPROXY	28	0.719	0.0003	0.976
1476	HE Qw1.5B	1D	112	0.747	0.0006	0.878
1477	HE Qw1.5B	DEFL	168	0.825	0.0021	0.952
1478	HE Qw1.5B	ANS	168	0.972	0.0006	0.978
1479	HE Qw1.5B	PH SINK	336	0.922	0.0007	0.998
1480	HE Qw1.5B	PH 0DTOT	336	0.921	0.0007	0.996
1481	HE Qw1.5B	PH 0D	672	0.949	0.0027	0.974
1482	HE Qw1.5B	PH ENT	336	0.933	0.0011	0.996
1483	HE Qw1.5B	LOOKBACK	336	0.999	0.0002	0.998
1484	HE Qw1.5B	HIDDEN	1536	1.000	0.0001	1.000
1485	HE Qw1.5B	NONTOPO	1962	1.000	0.0002	1.000
1486	HE Qw1.5B	0D+1D	168	0.815	0.0032	0.922
1487	HE Qw1.5B	0D+SINK	112	0.832	0.0015	0.924
1488	HE Qw1.5B	HIDDEN+0D	1592	1.000	0.0001	1.000
1489	HE Qw1.5B	LEX+0D	–	0.971	0.0004	0.986
1490	HE Qw1.5B	LEX+DEFL	–	0.956	0.0002	0.986
1491	HE Qw1.5B	LEX+SINK	–	0.968	0.0005	0.984
1492	HE Qw1.5B	LOGPROB+0D	62	0.994	0.0002	0.990
1493	HE Qw1.5B	NONTOPO+0D	2018	1.000	0.0000	1.000
1494	HE Qw1.5B	NONTOPO+DEFL	2130	1.000	0.0001	1.000
1495	HE Qw1.5B	PH SINK+PH 0DTOT	672	0.931	0.0024	0.998
1496	HE Qw1.5B	SINK+1D	168	0.826	0.0035	0.940
1497	HE Qw1.5B	SINK+ANS	224	0.975	0.0015	0.990
1498	HE Qw1.5B	SINK+DEFL	224	0.872	0.0028	0.962
1499	TrQA Qw3B	LEN	2	0.569	0.0014	–
1500	TrQA Qw3B	LEX	–	0.596	0.0039	–
1501	TrQA Qw3B	LOGPROB	6	0.839	0.0002	–
1502	TrQA Qw3B	LLMCHECK	36	0.688	0.0012	–
1503	TrQA Qw3B	ROWSTAT	72	0.724	0.0046	–
1504	TrQA Qw3B	SINK	72	0.717	0.0038	–
1505	TrQA Qw3B	0D	72	0.732	0.0038	–
1506	TrQA Qw3B	LEGACY MSTPROXY	36	0.691	0.0033	–
1507	TrQA Qw3B	1D	144	0.601	0.0031	–
1508	TrQA Qw3B	DEFL	216	0.684	0.0043	–
1509	TrQA Qw3B	ANS	216	0.719	0.0020	–
1510	TrQA Qw3B	PH SINK	576	0.820	0.0021	–
1511	TrQA Qw3B	PH 0DTOT	576	0.807	0.0012	–
	TrQA Qw3B	PH 0D	1152	0.807	0.0017	–
	TrQA Qw3B	PH ENT	576	0.810	0.0007	–
	TrQA Qw3B	LOOKBACK	576	0.861	0.0013	–
	TrQA Qw3B	HIDDEN	2048	0.841	0.0042	–
	TrQA Qw3B	NONTOPO	2738	0.870	0.0028	–
	TrQA Qw3B	0D+1D	216	0.729	0.0022	–
	TrQA Qw3B	0D+SINK	144	0.738	0.0050	–
	TrQA Qw3B	HIDDEN+0D	2120	0.843	0.0040	–
	TrQA Qw3B	LEX+0D	–	0.733	0.0011	–
	TrQA Qw3B	LEX+DEFL	–	0.690	0.0031	–
	TrQA Qw3B	LEX+SINK	–	0.724	0.0005	–

	Setting	Bank	dim	AUC	seed SD	pair acc.
1512						
1513						
1514	TrQA Qw3B	LOGPROB+0D	78	0.843	0.0022	—
1515	TrQA Qw3B	NONTOPO+0D	2810	0.870	0.0027	—
1516	TrQA Qw3B	NONTOPO+DEFL	2954	0.870	0.0029	—
1517	TrQA Qw3B	PH SINK+PH 0DTOT	1152	0.824	0.0017	—
1518	TrQA Qw3B	SINK+1D	216	0.718	0.0035	—
1519	TrQA Qw3B	SINK+ANS	288	0.752	0.0010	—
1520	TrQA Qw3B	SINK+DEFL	288	0.719	0.0038	—
1521	TrQA Qw1.5B	LEN	2	0.588	0.0012	—
1522	TrQA Qw1.5B	LEX	—	0.609	0.0033	—
1523	TrQA Qw1.5B	LOGPROB	6	0.823	0.0006	—
1524	TrQA Qw1.5B	LLMCHECK	28	0.676	0.0029	—
1525	TrQA Qw1.5B	ROWSTAT	56	0.720	0.0008	—
1526	TrQA Qw1.5B	SINK	56	0.707	0.0000	—
1527	TrQA Qw1.5B	0D	56	0.716	0.0004	—
1528	TrQA Qw1.5B	LEGACY MSTPROXY	28	0.685	0.0014	—
1529	TrQA Qw1.5B	1D	112	0.574	0.0024	—
1530	TrQA Qw1.5B	DEFL	168	0.655	0.0016	—
1531	TrQA Qw1.5B	ANS	168	0.675	0.0035	—
1532	TrQA Qw1.5B	PH SINK	336	0.767	0.0034	—
1533	TrQA Qw1.5B	PH 0DTOT	336	0.759	0.0051	—
1534	TrQA Qw1.5B	PH 0D	672	0.768	0.0013	—
1535	TrQA Qw1.5B	PH ENT	336	0.768	0.0025	—
1536	TrQA Qw1.5B	LOOKBACK	336	0.824	0.0017	—
1537	TrQA Qw1.5B	HIDDEN	1536	0.855	0.0031	—
1538	TrQA Qw1.5B	NONTOPO	1962	0.873	0.0023	—
1539	TrQA Qw1.5B	0D+1D	168	0.714	0.0015	—
1540	TrQA Qw1.5B	0D+SINK	112	0.717	0.0006	—
1541	TrQA Qw1.5B	HIDDEN+0D	1592	0.855	0.0032	—
1542	TrQA Qw1.5B	LEX+0D	—	0.721	0.0031	—
1543	TrQA Qw1.5B	LEX+DEFL	—	0.672	0.0047	—
1544	TrQA Qw1.5B	LEX+SINK	—	0.715	0.0036	—
1545	TrQA Qw1.5B	LOGPROB+0D	62	0.828	0.0007	—
1546	TrQA Qw1.5B	NONTOPO+0D	2018	0.873	0.0025	—
1547	TrQA Qw1.5B	NONTOPO+DEFL	2130	0.872	0.0024	—
1548	TrQA Qw1.5B	PH SINK+PH 0DTOT	672	0.772	0.0022	—
1549	TrQA Qw1.5B	SINK+1D	168	0.709	0.0028	—
1550	TrQA Qw1.5B	SINK+ANS	224	0.726	0.0059	—
1551	TrQA Qw1.5B	SINK+DEFL	224	0.719	0.0023	—
1552	TrQA Phi-3	LEN	2	0.581	0.0022	—
1553	TrQA Phi-3	LEX	—	0.635	0.0083	—
1554	TrQA Phi-3	LOGPROB	6	0.877	0.0002	—
1555	TrQA Phi-3	LLMCHECK	32	0.676	0.0018	—
1556	TrQA Phi-3	ROWSTAT	64	0.728	0.0006	—
1557	TrQA Phi-3	SINK	64	0.707	0.0018	—
1558	TrQA Phi-3	0D	64	0.710	0.0022	—
1559	TrQA Phi-3	LEGACY MSTPROXY	32	0.669	0.0002	—
1560	TrQA Phi-3	1D	128	0.546	0.0031	—
1561	TrQA Phi-3	DEFL	192	0.698	0.0021	—
1562	TrQA Phi-3	ANS	192	0.710	0.0037	—
1563	TrQA Phi-3	PH SINK	1024	0.826	0.0010	—
1564	TrQA Phi-3	PH 0DTOT	1024	0.810	0.0016	—
1565	TrQA Phi-3	PH 0D	2048	0.807	0.0024	—
	TrQA Phi-3	PH ENT	1024	0.817	0.0021	—
	TrQA Phi-3	LOOKBACK	1024	0.840	0.0016	—
	TrQA Phi-3	HIDDEN	3072	0.867	0.0035	—
	TrQA Phi-3	NONTOPO	4198	0.883	0.0030	—
	TrQA Phi-3	0D+1D	192	0.705	0.0011	—
	TrQA Phi-3	0D+SINK	128	0.713	0.0024	—
	TrQA Phi-3	HIDDEN+0D	3136	0.867	0.0037	—
	TrQA Phi-3	LEX+0D	—	0.721	0.0031	—
	TrQA Phi-3	LEX+DEFL	—	0.712	0.0022	—
	TrQA Phi-3	LEX+SINK	—	0.718	0.0044	—
	TrQA Phi-3	LOGPROB+0D	70	0.878	0.0013	—

	Setting	Bank	dim	AUC	seed SD	pair acc.
1566						
1567						
1568	TrQA Phi-3	NONTOPO+0D	4262	0.883	0.0030	—
1569	TrQA Phi-3	NONTOPO+DEFL	4390	0.885	0.0030	—
1570	TrQA Phi-3	PH SINK+PH 0DTOT	2048	0.828	0.0009	—
1571	TrQA Phi-3	SINK+1D	192	0.699	0.0012	—
1572	TrQA Phi-3	SINK+ANS	256	0.736	0.0046	—
1573	TrQA Phi-3	SINK+DEFL	256	0.736	0.0016	—
1574	TrQA TinyLl.	LEN	2	0.631	0.0005	—
1575	TrQA TinyLl.	LEX	—	0.740	0.0003	—
1576	TrQA TinyLl.	LOGPROB	6	0.724	0.0006	—
1577	TrQA TinyLl.	LLMCHECK	22	0.688	0.0010	—
1578	TrQA TinyLl.	ROWSTAT	44	0.752	0.0001	—
1579	TrQA TinyLl.	SINK	44	0.720	0.0020	—
1580	TrQA TinyLl.	0D	44	0.720	0.0020	—
1581	TrQA TinyLl.	LEGACY MSTPROXY	22	0.729	0.0006	—
1582	TrQA TinyLl.	1D	88	0.659	0.0007	—
1583	TrQA TinyLl.	DEFL	132	0.726	0.0024	—
1584	TrQA TinyLl.	ANS	132	0.701	0.0021	—
1585	TrQA TinyLl.	PH SINK	704	0.814	0.0016	—
1586	TrQA TinyLl.	PH 0DTOT	704	0.792	0.0018	—
1587	TrQA TinyLl.	PH 0D	1408	0.775	0.0027	—
1588	TrQA TinyLl.	PH ENT	704	0.807	0.0003	—
1589	TrQA TinyLl.	LOOKBACK	704	0.823	0.0024	—
1590	TrQA TinyLl.	HIDDEN	2048	0.827	0.0028	—
1591	TrQA TinyLl.	NONTOPO	2824	0.842	0.0031	—
1592	TrQA TinyLl.	0D+1D	132	0.718	0.0002	—
1593	TrQA TinyLl.	0D+SINK	88	0.724	0.0015	—
1594	TrQA TinyLl.	HIDDEN+0D	2092	0.828	0.0027	—
1595	TrQA TinyLl.	LEX+0D	—	0.764	0.0010	—
1596	TrQA TinyLl.	LEX+DEFL	—	0.757	0.0032	—
1597	TrQA TinyLl.	LEX+SINK	—	0.762	0.0021	—
1598	TrQA TinyLl.	LOGPROB+0D	50	0.762	0.0021	—
1599	TrQA TinyLl.	NONTOPO+0D	2868	0.842	0.0029	—
1600	TrQA TinyLl.	NONTOPO+DEFL	2956	0.842	0.0032	—
1601	TrQA TinyLl.	PH SINK+PH 0DTOT	1408	0.813	0.0016	—
1602	TrQA TinyLl.	SINK+1D	132	0.719	0.0013	—
1603	TrQA TinyLl.	SINK+ANS	176	0.725	0.0030	—
1604	TrQA TinyLl.	SINK+DEFL	176	0.734	0.0032	—
1605	TrQA Mis7B	LEN	2	0.635	0.0017	—
1606	TrQA Mis7B	LEX	—	0.677	0.0050	—
1607	TrQA Mis7B	LOGPROB	6	0.719	0.0005	—
1608	TrQA Mis7B	LLMCHECK	32	0.686	0.0019	—
1609	TrQA Mis7B	ROWSTAT	64	0.731	0.0016	—
1610	TrQA Mis7B	SINK	64	0.696	0.0017	—
1611	TrQA Mis7B	0D	64	0.699	0.0003	—
1612	TrQA Mis7B	LEGACY MSTPROXY	32	0.693	0.0017	—
1613	TrQA Mis7B	1D	128	0.611	0.0028	—
1614	TrQA Mis7B	DEFL	192	0.721	0.0024	—
1615	TrQA Mis7B	ANS	192	0.641	0.0033	—
1616	TrQA Mis7B	PH SINK	1024	0.812	0.0024	—
1617	TrQA Mis7B	PH 0DTOT	1024	0.808	0.0037	—
1618	TrQA Mis7B	PH 0D	2048	0.775	0.0013	—
1619	TrQA Mis7B	PH ENT	1024	0.812	0.0027	—
1620	TrQA Mis7B	LOOKBACK	1024	0.822	0.0035	—
1621	TrQA Mis7B	HIDDEN	4096	0.833	0.0007	—
1622	TrQA Mis7B	NONTOPO	5222	0.841	0.0010	—
1623	TrQA Mis7B	0D+1D	192	0.692	0.0035	—
1624	TrQA Mis7B	0D+SINK	128	0.698	0.0018	—
1625	TrQA Mis7B	HIDDEN+0D	4160	0.834	0.0012	—
1626	TrQA Mis7B	LEX+0D	—	0.723	0.0030	—
1627	TrQA Mis7B	LEX+DEFL	—	0.744	0.0057	—
1628	TrQA Mis7B	LEX+SINK	—	0.721	0.0027	—
1629	TrQA Mis7B	LOGPROB+0D	70	0.752	0.0006	—
1630	TrQA Mis7B	NONTOPO+0D	5286	0.841	0.0015	—

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Mis7B	NONTOPO+DEFL	5414	0.841	0.0011	—
TrQA Mis7B	PH SINK+PH 0DTOT	2048	0.814	0.0022	—
TrQA Mis7B	SINK+1D	192	0.686	0.0042	—
TrQA Mis7B	SINK+ANS	256	0.701	0.0025	—
TrQA Mis7B	SINK+DEFL	256	0.739	0.0019	—
TrQA Qw7B	LEN	2	0.595	0.0004	—
TrQA Qw7B	LEX	—	0.594	0.0029	—
TrQA Qw7B	LOGPROB	6	0.847	0.0003	—
TrQA Qw7B	LLMCHECK	28	0.725	0.0015	—
TrQA Qw7B	ROWSTAT	56	0.734	0.0016	—
TrQA Qw7B	SINK	56	0.714	0.0023	—
TrQA Qw7B	0D	56	0.668	0.0006	—
TrQA Qw7B	LEGACY MSTPROXY	28	0.672	0.0005	—
TrQA Qw7B	1D	112	0.582	0.0022	—
TrQA Qw7B	DEFL	168	0.672	0.0016	—
TrQA Qw7B	ANS	168	0.755	0.0026	—
TrQA Qw7B	PH SINK	784	0.801	0.0009	—
TrQA Qw7B	PH 0DTOT	784	0.822	0.0011	—
TrQA Qw7B	PH 0D	1568	0.798	0.0029	—
TrQA Qw7B	PH ENT	784	0.841	0.0021	—
TrQA Qw7B	LOOKBACK	784	0.879	0.0010	—
TrQA Qw7B	HIDDEN	3584	0.854	0.0007	—
TrQA Qw7B	NONTOPO	4458	0.881	0.0008	—
TrQA Qw7B	0D+1D	168	0.668	0.0021	—
TrQA Qw7B	0D+SINK	112	0.735	0.0008	—
TrQA Qw7B	HIDDEN+0D	3640	0.853	0.0008	—
TrQA Qw7B	LEX+0D	—	0.678	0.0043	—
TrQA Qw7B	LEX+DEFL	—	0.678	0.0022	—
TrQA Qw7B	LEX+SINK	—	0.728	0.0007	—
TrQA Qw7B	LOGPROB+0D	62	0.842	0.0012	—
TrQA Qw7B	NONTOPO+0D	4514	0.881	0.0007	—
TrQA Qw7B	NONTOPO+DEFL	4626	0.881	0.0007	—
TrQA Qw7B	PH SINK+PH 0DTOT	1568	0.832	0.0010	—
TrQA Qw7B	SINK+1D	168	0.718	0.0023	—
TrQA Qw7B	SINK+ANS	224	0.781	0.0035	—
TrQA Qw7B	SINK+DEFL	224	0.734	0.0021	—

Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TQA Qw3B	0D→SINK	0.736	0.737	+0.001	[-0.006, +0.009]	yes	0.90
TQA Qw3B	0D→LEGACY_MSTPROXY	0.736	0.713	-0.023	[-0.034, -0.012]	no	0.42
TQA Qw3B	SINK→0D+SINK	0.737	0.741	+0.003	[-0.003, +0.010]	yes	0.97
TQA Qw3B	SINK→SINK+DEFL	0.737	0.766	+0.029	[+0.017, +0.041]	no	0.30
TQA Qw3B	0D→0D+1D	0.736	0.769	+0.033	[+0.022, +0.045]	no	0.38
TQA Qw3B	SINK→SINK+1D	0.737	0.766	+0.028	[+0.017, +0.039]	no	0.46
TQA Qw3B	SINK→SINK+ANS	0.737	0.785	+0.047	[+0.033, +0.062]	no	0.03
TQA Qw3B	PH_0DTOT→PH_SINK	0.842	0.851	+0.009	[+0.002, +0.016]	no	0.93
TQA Qw3B	PH_SINK→PH_SINK+PH_0DTOT	0.851	0.860	+0.009	[+0.005, +0.012]	yes	1.00
TQA Qw3B	PH_0D→PH_ENT	0.853	0.857	+0.004	[-0.006, +0.014]	yes	0.57
TQA Qw3B	0D→LOGPROB	0.736	0.630	-0.106	[-0.129, -0.083]	no	0.00
TQA Qw3B	0D→HIDDEN	0.736	0.913	+0.177	[+0.163, +0.192]	no	0.03
TQA Qw3B	0D→LOOKBACK	0.736	0.897	+0.161	[+0.146, +0.175]	no	0.02
TQA Qw3B	0D→LLMCHECK	0.736	0.693	-0.043	[-0.059, -0.026]	no	0.00
TQA Qw3B	0D→LEX	0.736	0.820	+0.084	[+0.066, +0.102]	no	0.00
TQA Qw3B	0D→ROWSTAT	0.736	0.750	+0.014	[+0.002, +0.026]	no	0.30
TQA Qw3B	0D→LEN	0.736	0.529	-0.207	[-0.229, -0.185]	no	0.00
TQA Qw3B	NONTOPO→NONTOPO+0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO→NONTOPO+DEFL	0.926	0.928	+0.001	[-0.000, +0.003]	yes	1.00
TQA Qw3B	LEX→LEX+0D	0.820	0.840	+0.020	[+0.011, +0.030]	no	0.63
TQA Qw3B	LEX→LEX+SINK	0.820	0.843	+0.022	[+0.013, +0.033]	no	0.59
TQA Qw3B	LEX→LEX+DEFL	0.820	0.830	+0.010	[-0.003, +0.023]	no	0.19
TQA Qw3B	LOGPROB→LOGPROB+0D	0.630	0.776	+0.146	[+0.127, +0.165]	no	0.00
TQA Qw3B	HIDDEN→HIDDEN+0D	0.913	0.913	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO→NONTOPO(+)+0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO→NONTOPO(+)+DEFL	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO→NONTOPO(+)+SINK	0.926	0.926	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw3B	HIDDEN→HIDDEN(+)+0D	0.913	0.914	+0.001	[-0.000, +0.002]	yes	1.00
TQA Qw3B	LOGPROB→LOGPROB(+)+0D	0.630	0.757	+0.128	[+0.109, +0.147]	no	0.00
TQA Qw3B	SINK→SINK(+)+0D	0.737	0.739	+0.002	[-0.001, +0.006]	yes	1.00
TQA Qw1.5B	0D→SINK	0.726	0.723	-0.003	[-0.012, -0.006]	yes	0.68
TQA Qw1.5B	0D→LEGACY_MSTPROXY	0.726	0.698	-0.028	[-0.040, -0.016]	no	0.36
TQA Qw1.5B	SINK→0D+SINK	0.723	0.735	+0.012	[+0.005, +0.020]	no	0.91

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡ _{.015}	power
1674	TQA Qw1.5B	SINK → SINK+DEFL	0.723	0.757	+0.034	[+0.022, +0.045]	no	0.35
1675	TQA Qw1.5B	0D → 0D+1D	0.726	0.718	-0.008	[-0.014, -0.002]	yes	0.99
1676	TQA Qw1.5B	SINK → SINK+1D	0.723	0.715	-0.008	[-0.014, -0.002]	yes	0.99
1677	TQA Qw1.5B	SINK → SINK+ANS	0.723	0.761	+0.038	[+0.024, +0.051]	no	0.11
1678	TQA Qw1.5B	PH_0DTOT → PH_SINK	0.819	0.830	+0.011	[+0.003, +0.018]	no	0.90
1679	TQA Qw1.5B	PH_SINK → PH_SINK+PH_0DTOT	0.830	0.841	+0.011	[+0.007, +0.014]	yes	1.00
1679	TQA Qw1.5B	PH_0D → PH_ENT	0.838	0.833	-0.004	[-0.016, +0.007]	no	0.44
1680	TQA Qw1.5B	0D → LOGPROB	0.726	0.638	-0.088	[-0.108, -0.067]	no	0.00
1680	TQA Qw1.5B	0D → HIDDEN	0.726	0.916	+0.190	[+0.175, +0.204]	no	0.05
1681	TQA Qw1.5B	0D → LOOKBACK	0.726	0.851	+0.125	[+0.109, +0.140]	no	0.00
1681	TQA Qw1.5B	0D → LLMCHECK	0.726	0.666	-0.060	[-0.076, -0.044]	no	0.00
1682	TQA Qw1.5B	0D → LEX	0.726	0.820	+0.094	[+0.076, +0.112]	no	0.00
1682	TQA Qw1.5B	0D → ROWSTAT	0.726	0.738	+0.012	[-0.000, +0.025]	no	0.27
1683	TQA Qw1.5B	0D → LEN	0.726	0.529	-0.197	[-0.218, -0.176]	no	0.00
1683	TQA Qw1.5B	NONTOPO → NONTOPO+0D	0.917	0.917	+0.000	[-0.000, +0.001]	yes	1.00
1684	TQA Qw1.5B	NONTOPO → NONTOPO+DEFL	0.917	0.917	+0.000	[-0.001, +0.001]	yes	1.00
1685	TQA Qw1.5B	LEX → LEX+0D	0.820	0.835	+0.015	[+0.006, +0.024]	no	0.71
1685	TQA Qw1.5B	LEX → LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.022]	no	0.73
1686	TQA Qw1.5B	LEX → LEX+DEFL	0.820	0.825	+0.005	[-0.006, +0.016]	no	0.45
1686	TQA Qw1.5B	LOGPROB → LOGPROB+0D	0.638	0.740	+0.101	[+0.086, +0.116]	no	0.00
1687	TQA Qw1.5B	HIDDEN → HIDDEN+0D	0.916	0.916	+0.001	[-0.000, +0.002]	yes	1.00
1688	TQA Qw1.5B	NONTOPO → NONTOPO(+)0D	0.916	0.916	+0.000	[-0.001, +0.001]	yes	1.00
1688	TQA Qw1.5B	NONTOPO → NONTOPO(+)DEFL	0.916	0.916	-0.000	[-0.001, +0.001]	yes	1.00
1689	TQA Qw1.5B	NONTOPO → NONTOPO(+)SINK	0.916	0.916	-0.000	[-0.001, +0.000]	yes	1.00
1689	TQA Qw1.5B	HIDDEN → HIDDEN(+)0D	0.915	0.915	+0.000	[-0.001, +0.001]	yes	1.00
1690	TQA Qw1.5B	LOGPROB → LOGPROB(+)0D	0.638	0.737	+0.100	[+0.084, +0.115]	no	0.00
1691	TQA Qw1.5B	SINK → SINK(+)0D	0.722	0.730	+0.007	[+0.002, +0.012]	yes	1.00
1691	TQA Phi-3	0D → SINK	0.646	0.651	+0.005	[-0.003, +0.012]	yes	0.90
1692	TQA Phi-3	0D → LEGACY_MSTPROXY	0.646	0.613	-0.033	[-0.046, -0.020]	no	0.20
1693	TQA Phi-3	SINK → 0D+SINK	0.651	0.657	+0.006	[-0.000, +0.012]	yes	0.99
1693	TQA Phi-3	SINK → SINK+DEFL	0.651	0.684	+0.033	[+0.020, +0.048]	no	0.10
1694	TQA Phi-3	0D → 0D+1D	0.646	0.643	-0.003	[-0.006, +0.001]	yes	1.00
1694	TQA Phi-3	SINK → SINK+1D	0.651	0.651	+0.000	[-0.004, +0.005]	yes	1.00
1695	TQA Phi-3	SINK → SINK+ANS	0.651	0.675	+0.024	[+0.007, +0.039]	no	0.00
1695	TQA Phi-3	PH_0DTOT → PH_SINK	0.845	0.871	+0.026	[+0.020, +0.031]	no	1.00
1696	TQA Phi-3	PH_SINK → PH_SINK+PH_0DTOT	0.871	0.871	-0.000	[-0.001, +0.001]	yes	1.00
1696	TQA Phi-3	PH_0D → PH_ENT	0.868	0.864	-0.004	[-0.012, +0.004]	yes	0.87
1697	TQA Phi-3	0D → LOGPROB	0.646	0.578	-0.068	[-0.092, -0.044]	no	0.00
1697	TQA Phi-3	0D → HIDDEN	0.646	0.933	+0.287	[+0.272, +0.302]	no	0.00
1698	TQA Phi-3	0D → LOOKBACK	0.646	0.892	+0.246	[+0.230, +0.262]	no	0.00
1699	TQA Phi-3	0D → LLMCHECK	0.646	0.653	+0.007	[-0.010, +0.024]	no	0.00
1699	TQA Phi-3	0D → LEX	0.646	0.820	+0.174	[+0.154, +0.193]	no	0.00
1700	TQA Phi-3	0D → ROWSTAT	0.646	0.661	+0.015	[+0.002, +0.029]	no	0.12
1700	TQA Phi-3	0D → LEN	0.646	0.522	-0.124	[-0.145, -0.102]	no	0.00
1701	TQA Phi-3	NONTOPO → NONTOPO+0D	0.936	0.936	+0.000	[-0.000, +0.001]	yes	1.00
1702	TQA Phi-3	NONTOPO → NONTOPO+DEFL	0.936	0.937	+0.001	[-0.000, +0.002]	yes	1.00
1702	TQA Phi-3	LEX → LEX+0D	0.820	0.821	+0.001	[-0.007, +0.009]	yes	0.85
1703	TQA Phi-3	LEX → LEX+SINK	0.820	0.820	-0.000	[-0.008, +0.007]	yes	0.89
1703	TQA Phi-3	LEX → LEX+DEFL	0.820	0.802	-0.018	[-0.028, -0.007]	no	0.52
1704	TQA Phi-3	LOGPROB → LOGPROB+0D	0.578	0.655	+0.077	[+0.058, +0.096]	no	0.00
1705	TQA Phi-3	HIDDEN → HIDDEN+0D	0.933	0.933	+0.000	[-0.001, +0.001]	yes	1.00
1705	TQA Phi-3	NONTOPO → NONTOPO(+)0D	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
1706	TQA Phi-3	NONTOPO → NONTOPO(+)DEFL	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
1706	TQA Phi-3	NONTOPO → NONTOPO(+)SINK	0.936	0.936	-0.000	[-0.000, +0.000]	yes	1.00
1707	TQA Phi-3	HIDDEN → HIDDEN(+)0D	0.932	0.932	-0.000	[-0.000, -0.000]	yes	1.00
1707	TQA Phi-3	LOGPROB → LOGPROB(+)0D	0.577	0.650	+0.073	[+0.055, +0.091]	no	0.00
1708	TQA Phi-3	SINK → SINK(+)0D	0.650	0.651	+0.000	[-0.003, +0.003]	yes	1.00
1709	TQA TinyLl.	0D → SINK	0.654	0.643	-0.010	[-0.021, +0.000]	no	0.51
1709	TQA TinyLl.	0D → LEGACY_MSTPROXY	0.654	0.627	-0.027	[-0.037, -0.016]	no	0.46
1710	TQA TinyLl.	SINK → 0D+SINK	0.643	0.663	+0.020	[+0.011, +0.029]	no	0.73
1710	TQA TinyLl.	SINK → SINK+DEFL	0.643	0.688	+0.045	[+0.032, +0.057]	no	0.28
1711	TQA TinyLl.	0D → 0D+1D	0.654	0.660	+0.007	[-0.000, +0.013]	yes	0.95
1712	TQA TinyLl.	SINK → SINK+1D	0.643	0.651	+0.008	[+0.001, +0.015]	no	0.93
1712	TQA TinyLl.	SINK → SINK+ANS	0.643	0.685	+0.041	[+0.025, +0.058]	no	0.00
1713	TQA TinyLl.	PH_0DTOT → PH_SINK	0.804	0.808	+0.004	[-0.002, +0.011]	yes	0.97
1713	TQA TinyLl.	PH_SINK → PH_SINK+PH_0DTOT	0.808	0.810	+0.003	[+0.000, +0.005]	yes	1.00
1714	TQA TinyLl.	PH_0D → PH_ENT	0.806	0.817	+0.011	[+0.000, +0.022]	no	0.50
1715	TQA TinyLl.	0D → LOGPROB	0.654	0.637	-0.017	[-0.039, +0.004]	no	0.00
1715	TQA TinyLl.	0D → HIDDEN	0.654	0.883	+0.229	[+0.214, +0.244]	no	0.00
1716	TQA TinyLl.	0D → LOOKBACK	0.654	0.850	+0.197	[+0.182, +0.212]	no	0.00
1716	TQA TinyLl.	0D → LLMCHECK	0.654	0.614	-0.039	[-0.055, -0.024]	no	0.00
1717	TQA TinyLl.	0D → LEX	0.654	0.820	+0.166	[+0.149, +0.184]	no	0.00
1717	TQA TinyLl.	0D → ROWSTAT	0.654	0.661	+0.007	[-0.004, +0.019]	no	0.37
1718	TQA TinyLl.	0D → LEN	0.654	0.523	-0.131	[-0.152, -0.110]	no	0.00
1719	TQA TinyLl.	NONTOPO → NONTOPO+0D	0.885	0.885	+0.000	[-0.000, +0.000]	yes	1.00
1719	TQA TinyLl.	NONTOPO → NONTOPO+DEFL	0.885	0.884	-0.001	[-0.002, +0.000]	yes	1.00
1720	TQA TinyLl.	LEX → LEX+0D	0.820	0.830	+0.010	[+0.003, +0.016]	no	0.97
1720	TQA TinyLl.	LEX → LEX+SINK	0.820	0.831	+0.011	[+0.005, +0.017]	no	0.98
1721	TQA TinyLl.	LEX → LEX+DEFL	0.820	0.826	+0.005	[-0.003, +0.014]	yes	0.75
1722	TQA TinyLl.	LOGPROB → LOGPROB+0D	0.637	0.683	+0.046	[+0.033, +0.060]	no	0.13
1722	TQA TinyLl.	HIDDEN → HIDDEN+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
1723	TQA TinyLl.	NONTOPO → NONTOPO(+)0D	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
1723	TQA TinyLl.	NONTOPO → NONTOPO(+)DEFL	0.884	0.884	-0.000	[-0.001, +0.000]	yes	1.00
1724	TQA TinyLl.	NONTOPO → NONTOPO(+)SINK	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
1724	TQA TinyLl.	HIDDEN → HIDDEN(+)0D	0.883	0.882	-0.000	[-0.000, +0.000]	yes	1.00
1725	TQA TinyLl.	LOGPROB → LOGPROB(+)0D	0.636	0.678	+0.043	[+0.030, +0.055]	no	0.25
1725	TQA TinyLl.	SINK → SINK(+)0D	0.643	0.657	+0.014	[+0.007, +0.021]	no	0.96
1726	TQA SmoLLM	0D → SINK	0.583	0.611	+0.028	[+0.009, +0.048]	no	0.00
1727	TQA SmoLLM	0D → LEGACY_MSTPROXY	0.583	0.585	+0.002	[-0.003, +0.006]	yes	1.00
1727	TQA SmoLLM	SINK → 0D+SINK	0.611	0.641	+0.030	[+0.021, +0.039]	no	0.66

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡ _{.015}	power
1782	TQA Qw7B	SINK → SINK+DEFL	0.749	0.788	+0.038	[+0.029, +0.048]	no	0.64
1783	TQA Qw7B	0D → 0D+1D	0.679	0.701	+0.021	[+0.011, +0.032]	no	0.52
1784	TQA Qw7B	SINK → SINK+1D	0.749	0.767	+0.017	[+0.010, +0.025]	no	0.89
1785	TQA Qw7B	SINK → SINK+ANS	0.749	0.794	+0.045	[+0.030, +0.060]	no	0.02
1786	TQA Qw7B	PH_0DTOT → PH_SINK	0.872	0.821	-0.051	[-0.062, -0.040]	no	0.45
1787	TQA Qw7B	PH_SINK → PH_SINK+PH_0DTOT	0.821	0.881	+0.060	[+0.051, +0.069]	no	0.71
1787	TQA Qw7B	PH_0D → PH_ENT	0.850	0.887	+0.037	[+0.027, +0.048]	no	0.52
1788	TQA Qw7B	0D → LOGPROB	0.679	0.674	-0.006	[-0.028, +0.017]	no	0.00
1788	TQA Qw7B	0D → HIDDEN	0.679	0.933	+0.254	[+0.240, +0.268]	no	0.09
1789	TQA Qw7B	0D → LOOKBACK	0.679	0.922	+0.242	[+0.229, +0.256]	no	0.17
1789	TQA Qw7B	0D → LLMCHECK	0.679	0.667	-0.012	[-0.027, +0.004]	no	0.00
1790	TQA Qw7B	0D → LEX	0.679	0.820	+0.141	[+0.122, +0.159]	no	0.00
1790	TQA Qw7B	0D → ROWSTAT	0.679	0.736	+0.057	[+0.048, +0.066]	no	0.74
1791	TQA Qw7B	0D → LEN	0.679	0.529	-0.151	[-0.171, -0.130]	no	0.00
1791	TQA Qw7B	NONTOPO → NONTOPO+0D	0.950	0.949	-0.000	[-0.000, +0.000]	yes	1.00
1792	TQA Qw7B	NONTOPO → NONTOPO+DEFL	0.950	0.949	-0.000	[-0.001, +0.000]	yes	1.00
1793	TQA Qw7B	LEX → LEX+0D	0.820	0.834	+0.014	[+0.007, +0.021]	no	0.94
1793	TQA Qw7B	LEX → LEX+SINK	0.820	0.848	+0.028	[+0.016, +0.040]	no	0.29
1794	TQA Qw7B	LEX → LEX+DEFL	0.820	0.828	+0.008	[-0.001, +0.016]	no	0.78
1794	TQA Qw7B	LOGPROB → LOGPROB+0D	0.674	0.727	+0.053	[+0.042, +0.065]	no	0.40
1795	TQA Qw7B	HIDDEN → HIDDEN+0D	0.933	0.933	-0.000	[-0.001, +0.000]	yes	1.00
1796	TQA Qw7B	NONTOPO → NONTOPO(+)0D	0.949	0.949	-0.000	[-0.000, +0.000]	yes	1.00
1796	TQA Qw7B	NONTOPO → NONTOPO(+)DEFL	0.949	0.949	+0.000	[-0.000, +0.000]	yes	1.00
1797	TQA Qw7B	NONTOPO → NONTOPO(+)SINK	0.949	0.949	+0.000	[-0.001, +0.001]	yes	1.00
1797	TQA Qw7B	HIDDEN → HIDDEN(+)0D	0.933	0.933	+0.000	[-0.000, +0.001]	yes	1.00
1798	TQA Qw7B	LOGPROB → LOGPROB(+)0D	0.673	0.725	+0.052	[+0.039, +0.064]	no	0.23
1799	TQA Qw7B	SINK → SINK(+)0D	0.749	0.775	+0.026	[+0.018, +0.034]	no	0.88
1799	HE Qw3B	0D → SINK	0.811	0.829	+0.018	[+0.008, +0.028]	no	0.64
1800	HE Qw3B	0D → LEGACY_MSTPROXY	0.811	0.731	-0.080	[-0.091, -0.069]	no	0.46
1800	HE Qw3B	SINK → 0D+SINK	0.829	0.857	+0.028	[+0.022, +0.035]	no	0.97
1801	HE Qw3B	SINK → SINK+DEFL	0.829	0.887	+0.058	[+0.049, +0.067]	no	0.73
1801	HE Qw3B	0D → 0D+1D	0.811	0.826	+0.015	[+0.009, +0.023]	no	0.94
1802	HE Qw3B	SINK → SINK+1D	0.829	0.843	+0.014	[+0.007, +0.021]	no	0.97
1803	HE Qw3B	SINK → SINK+ANS	0.829	0.978	+0.150	[+0.139, +0.161]	no	0.44
1803	HE Qw3B	PH_0DTOT → PH_SINK	0.958	0.958	+0.000	[-0.005, +0.006]	yes	1.00
1804	HE Qw3B	PH_SINK → PH_SINK+PH_0DTOT	0.958	0.962	+0.004	[+0.003, +0.005]	yes	1.00
1804	HE Qw3B	PH_0D → PH_ENT	0.971	0.964	-0.006	[-0.012, -0.000]	yes	0.99
1805	HE Qw3B	0D → LOGPROB	0.811	0.993	+0.182	[+0.170, +0.194]	no	0.33
1805	HE Qw3B	0D → HIDDEN	0.811	1.000	+0.189	[+0.177, +0.201]	no	0.32
1806	HE Qw3B	0D → LOOKBACK	0.811	0.998	+0.187	[+0.175, +0.199]	no	0.32
1807	HE Qw3B	0D → LLMCHECK	0.811	0.745	-0.066	[-0.078, -0.054]	no	0.27
1807	HE Qw3B	0D → LEX	0.811	0.981	+0.170	[+0.159, +0.182]	no	0.37
1808	HE Qw3B	0D → ROWSTAT	0.811	0.829	+0.018	[+0.007, +0.030]	no	0.35
1808	HE Qw3B	0D → LEN	0.811	0.962	+0.151	[+0.140, +0.163]	no	0.36
1809	HE Qw3B	NONTOPO → NONTOPO+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1810	HE Qw3B	NONTOPO → NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1810	HE Qw3B	LEX → LEX+0D	0.981	0.979	-0.002	[-0.005, +0.000]	yes	1.00
1811	HE Qw3B	LEX → LEX+SINK	0.981	0.980	-0.001	[-0.003, +0.001]	yes	1.00
1811	HE Qw3B	LEX → LEX+DEFL	0.981	0.971	-0.010	[-0.014, -0.007]	yes	1.00
1812	HE Qw3B	LOGPROB → LOGPROB+0D	0.993	0.993	+0.000	[-0.001, +0.001]	yes	1.00
1812	HE Qw3B	HIDDEN → HIDDEN+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1813	HE Qw3B	NONTOPO → NONTOPO(+)0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1814	HE Qw3B	NONTOPO → NONTOPO(+)DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1814	HE Qw3B	NONTOPO → NONTOPO(+)SINK	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1815	HE Qw3B	HIDDEN → HIDDEN(+)0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1815	HE Qw3B	LOGPROB → LOGPROB(+)0D	0.993	0.993	+0.001	[+0.000, +0.001]	yes	1.00
1816	HE Qw3B	SINK → SINK(+)0D	0.828	0.839	+0.011	[+0.007, +0.015]	yes	1.00
1817	HE Qw1.5B	0D → SINK	0.789	0.820	+0.031	[+0.017, +0.045]	no	0.08
1817	HE Qw1.5B	0D → LEGACY_MSTPROXY	0.789	0.719	-0.070	[-0.086, -0.053]	no	0.00
1818	HE Qw1.5B	SINK → 0D+SINK	0.820	0.832	+0.012	[+0.005, +0.019]	no	0.91
1819	HE Qw1.5B	SINK → SINK+DEFL	0.820	0.872	+0.052	[+0.038, +0.065]	no	0.17
1819	HE Qw1.5B	0D → 0D+1D	0.789	0.815	+0.026	[+0.013, +0.040]	no	0.17
1820	HE Qw1.5B	SINK → SINK+1D	0.820	0.826	+0.006	[-0.005, +0.017]	no	0.45
1820	HE Qw1.5B	SINK → SINK+ANS	0.820	0.975	+0.155	[+0.138, +0.172]	no	0.00
1821	HE Qw1.5B	PH_0DTOT → PH_SINK	0.921	0.922	+0.002	[-0.009, +0.013]	yes	0.46
1821	HE Qw1.5B	PH_SINK → PH_SINK+PH_0DTOT	0.922	0.931	+0.008	[+0.006, +0.011]	yes	1.00
1822	HE Qw1.5B	PH_0D → PH_ENT	0.949	0.933	-0.016	[-0.028, -0.004]	no	0.33
1823	HE Qw1.5B	0D → LOGPROB	0.789	0.995	+0.205	[+0.188, +0.222]	no	0.00
1823	HE Qw1.5B	0D → HIDDEN	0.789	1.000	+0.210	[+0.193, +0.227]	no	0.00
1824	HE Qw1.5B	0D → LOOKBACK	0.789	0.999	+0.210	[+0.193, +0.226]	no	0.00
1824	HE Qw1.5B	0D → LLMCHECK	0.789	0.791	+0.002	[-0.017, +0.021]	no	0.00
1825	HE Qw1.5B	0D → LEX	0.789	0.979	+0.190	[+0.172, +0.207]	no	0.00
1825	HE Qw1.5B	0D → ROWSTAT	0.789	0.788	-0.001	[-0.018, +0.016]	no	0.00
1826	HE Qw1.5B	0D → LEN	0.789	0.967	+0.178	[+0.160, +0.196]	no	0.00
1827	HE Qw1.5B	NONTOPO → NONTOPO+0D	1.000	1.000	+0.000	[+0.000, +0.000]	yes	1.00
1827	HE Qw1.5B	NONTOPO → NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1828	HE Qw1.5B	LEX → LEX+0D	0.979	0.971	-0.008	[-0.014, -0.002]	yes	0.99
1828	HE Qw1.5B	LEX → LEX+SINK	0.979	0.968	-0.011	[-0.017, -0.005]	no	0.98
1829	HE Qw1.5B	LEX → LEX+DEFL	0.979	0.956	-0.022	[-0.030, -0.016]	no	0.94
1830	HE Qw1.5B	LOGPROB → LOGPROB+0D	0.995	0.994	-0.001	[-0.003, +0.001]	yes	1.00
1830	HE Qw1.5B	HIDDEN → HIDDEN+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1831	HE Qw1.5B	NONTOPO → NONTOPO(+)0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1831	HE Qw1.5B	NONTOPO → NONTOPO(+)DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1832	HE Qw1.5B	NONTOPO → NONTOPO(+)SINK	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1832	HE Qw1.5B	HIDDEN → HIDDEN(+)0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1833	HE Qw1.5B	LOGPROB → LOGPROB(+)0D	0.994	0.994	-0.000	[-0.001, +0.001]	yes	1.00
1833	HE Qw1.5B	SINK → SINK(+)0D	0.819	0.823	+0.004	[-0.000, +0.009]	yes	1.00
1834	TrQA Qw3B	0D → SINK	0.732	0.717	-0.015	[-0.026, -0.004]	no	0.48
1834	TrQA Qw3B	0D → LEGACY_MSTPROXY	0.732	0.691	-0.041	[-0.057, -0.025]	no	0.00
1835	TrQA Qw3B	SINK → 0D+SINK	0.717	0.738	+0.021	[+0.012, +0.030]	no	0.76

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TrQA Qw3B	SINK $\rightarrow$ SINK+DEFL	0.717	0.719	+0.002	[-0.012, +0.015]	no	0.13
TrQA Qw3B	0D $\rightarrow$ 0D+1D	0.732	0.729	-0.003	[-0.012, +0.006]	yes	0.74
TrQA Qw3B	SINK $\rightarrow$ SINK+1D	0.717	0.718	+0.001	[-0.009, +0.010]	yes	0.65
TrQA Qw3B	SINK $\rightarrow$ SINK+ANS	0.717	0.752	+0.035	[+0.019, +0.051]	no	0.00
TrQA Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.807	0.820	+0.012	[+0.004, +0.021]	no	0.82
TrQA Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.820	0.824	+0.004	[+0.002, +0.006]	yes	1.00
TrQA Qw3B	PH_0D $\rightarrow$ PH_ENT	0.807	0.810	+0.003	[-0.011, +0.017]	no	0.08
TrQA Qw3B	0D $\rightarrow$ LOGPROB	0.732	0.839	+0.107	[+0.088, +0.125]	no	0.00
TrQA Qw3B	0D $\rightarrow$ HIDDEN	0.732	0.841	+0.109	[+0.090, +0.129]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LOOKBACK	0.732	0.861	+0.129	[+0.111, +0.147]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LLMCHECK	0.732	0.688	-0.044	[-0.065, -0.023]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEX	0.732	0.596	-0.137	[-0.163, -0.110]	no	0.00
TrQA Qw3B	0D $\rightarrow$ ROWSTAT	0.732	0.724	-0.009	[-0.025, +0.008]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEN	0.732	0.569	-0.163	[-0.189, -0.137]	no	0.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.870	0.870	+0.001	[-0.000, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.870	0.870	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	LEX $\rightarrow$ LEX+0D	0.596	0.733	+0.137	[+0.117, +0.158]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.596	0.724	+0.128	[+0.108, +0.149]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.596	0.690	+0.095	[+0.073, +0.116]	no	0.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.839	0.843	+0.004	[-0.002, +0.011]	yes	0.95
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.841	0.843	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.869	0.871	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.869	0.870	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.869	0.871	+0.001	[-0.001, +0.004]	yes	1.00
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.841	0.849	+0.008	[+0.003, +0.012]	yes	1.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.838	0.844	+0.006	[+0.002, +0.010]	yes	1.00
TrQA Qw3B	SINK $\rightarrow$ SINK(+)0D	0.716	0.733	+0.017	[+0.010, +0.025]	no	0.89
TrQA Qw1.5B	0D $\rightarrow$ SINK	0.716	0.707	-0.009	[-0.020, +0.002]	no	0.43
TrQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.716	0.685	-0.031	[-0.046, -0.015]	no	0.00
TrQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.707	0.717	+0.010	[+0.001, +0.020]	no	0.68
TrQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.707	0.719	+0.012	[-0.003, +0.027]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.716	0.714	-0.002	[-0.011, +0.008]	yes	0.65
TrQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.707	0.709	+0.002	[-0.008, +0.013]	yes	0.49
TrQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.707	0.726	+0.019	[+0.001, +0.036]	no	0.00
TrQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.759	0.767	+0.008	[-0.003, +0.018]	no	0.53
TrQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.767	0.772	+0.004	[+0.001, +0.008]	yes	1.00
TrQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.768	0.768	-0.000	[-0.017, +0.017]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.716	0.823	+0.107	[+0.085, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.716	0.855	+0.140	[+0.119, +0.160]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.716	0.824	+0.108	[+0.088, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.716	0.676	-0.039	[-0.062, -0.016]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEX	0.716	0.609	-0.106	[-0.136, -0.078]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.716	0.720	+0.005	[-0.011, +0.020]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEN	0.716	0.588	-0.128	[-0.154, -0.102]	no	0.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.873	0.873	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.873	0.872	-0.001	[-0.002, +0.001]	yes	1.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.609	0.721	+0.112	[+0.091, +0.134]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.609	0.715	+0.106	[+0.085, +0.127]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.609	0.672	+0.063	[+0.040, +0.085]	no	0.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.823	0.828	+0.006	[-0.001, +0.013]	yes	0.92
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.855	0.855	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.872	0.873	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.855	0.857	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.822	0.828	+0.006	[+0.001, +0.010]	yes	1.00
TrQA Qw1.5B	SINK $\rightarrow$ SINK(+)0D	0.706	0.717	+0.011	[+0.003, +0.018]	no	0.93
TrQA Phi-3	0D $\rightarrow$ SINK	0.710	0.707	-0.004	[-0.010, +0.002]	yes	0.98
TrQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.710	0.669	-0.041	[-0.057, -0.025]	no	0.00
TrQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.707	0.713	+0.006	[+0.000, +0.012]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.707	0.736	+0.029	[+0.013, +0.045]	no	0.00
TrQA Phi-3	0D $\rightarrow$ 0D+1D	0.710	0.705	-0.005	[-0.011, +0.000]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+1D	0.707	0.699	-0.008	[-0.013, -0.002]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.707	0.736	+0.029	[+0.012, +0.046]	no	0.00
TrQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.810	0.826	+0.016	[+0.009, +0.022]	no	0.96
TrQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.826	0.828	+0.002	[-0.001, +0.004]	yes	1.00
TrQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.807	0.817	+0.010	[-0.002, +0.022]	no	0.31
TrQA Phi-3	0D $\rightarrow$ LOGPROB	0.710	0.877	+0.167	[+0.148, +0.186]	no	0.00
TrQA Phi-3	0D $\rightarrow$ HIDDEN	0.710	0.867	+0.157	[+0.137, +0.176]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LOOKBACK	0.710	0.840	+0.129	[+0.109, +0.150]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LLMCHECK	0.710	0.676	-0.034	[-0.055, -0.014]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEX	0.710	0.635	-0.076	[-0.102, -0.050]	no	0.00
TrQA Phi-3	0D $\rightarrow$ ROWSTAT	0.710	0.728	+0.018	[+0.002, +0.034]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEN	0.710	0.581	-0.130	[-0.156, -0.104]	no	0.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.883	0.885	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Phi-3	LEX $\rightarrow$ LEX+0D	0.635	0.721	+0.086	[+0.069, +0.104]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.635	0.718	+0.083	[+0.066, +0.101]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.635	0.712	+0.077	[+0.057, +0.098]	no	0.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.877	0.878	+0.000	[-0.002, +0.002]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.867	0.867	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.883	0.884	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.883	0.884	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.883	0.883	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.867	0.869	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.877	0.878	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK(+)0D	0.706	0.710	+0.004	[-0.001, +0.009]	yes	1.00
TrQA TinyLL	0D $\rightarrow$ SINK	0.720	0.720	+0.001	[-0.010, +0.012]	yes	0.40
TrQA TinyLL	0D $\rightarrow$ LEGACY_MSTPROXY	0.720	0.729	+0.010	[-0.000, +0.020]	no	0.55
TrQA TinyLL	SINK $\rightarrow$ 0D+SINK	0.720	0.724	+0.004	[-0.003, +0.011]	yes	0.93

	Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
1890								
1891	TrQA TinyLL	SINK $\rightarrow$ SINK+DEFL	0.720	0.734	+0.014	[-0.000, +0.028]	no	0.10
1892	TrQA TinyLL	0D $\rightarrow$ 0D+1D	0.720	0.718	-0.002	[-0.011, +0.008]	yes	0.60
1893	TrQA TinyLL	SINK $\rightarrow$ SINK+1D	0.720	0.719	-0.001	[-0.010, +0.008]	yes	0.73
1894	TrQA TinyLL	SINK $\rightarrow$ SINK+ANS	0.720	0.725	+0.005	[-0.011, +0.021]	no	0.00
1894	TrQA TinyLL	PH_0DTOT $\rightarrow$ PH_SINK	0.792	0.814	+0.022	[+0.015, +0.029]	no	0.94
1894	TrQA TinyLL	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.814	0.813	-0.001	[-0.002, +0.000]	yes	1.00
1895	TrQA TinyLL	PH_0D $\rightarrow$ PH_ENT	0.775	0.807	+0.032	[+0.017, +0.047]	no	0.01
1896	TrQA TinyLL	0D $\rightarrow$ LOGPROB	0.720	0.724	+0.004	[-0.019, +0.027]	no	0.00
1896	TrQA TinyLL	0D $\rightarrow$ HIDDEN	0.720	0.827	+0.108	[+0.088, +0.128]	no	0.00
1897	TrQA TinyLL	0D $\rightarrow$ LOOKBACK	0.720	0.823	+0.104	[+0.087, +0.121]	no	0.00
1897	TrQA TinyLL	0D $\rightarrow$ LLMCHECK	0.720	0.688	-0.032	[-0.050, -0.013]	no	0.00
1898	TrQA TinyLL	0D $\rightarrow$ LEX	0.720	0.740	+0.021	[-0.003, +0.044]	no	0.00
1898	TrQA TinyLL	0D $\rightarrow$ ROWSTAT	0.720	0.752	+0.032	[+0.020, +0.045]	no	0.26
1899	TrQA TinyLL	0D $\rightarrow$ LEN	0.720	0.631	-0.089	[-0.109, -0.068]	no	0.00
1899	TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+0D	0.842	0.842	-0.000	[-0.001, +0.001]	yes	1.00
1900	TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.842	0.842	+0.000	[-0.003, +0.003]	yes	1.00
1901	TrQA TinyLL	LEX $\rightarrow$ LEX+0D	0.740	0.764	+0.024	[+0.012, +0.036]	no	0.33
1901	TrQA TinyLL	LEX $\rightarrow$ LEX+SINK	0.740	0.762	+0.021	[+0.010, +0.032]	no	0.45
1902	TrQA TinyLL	LEX $\rightarrow$ LEX+DEFL	0.740	0.757	+0.016	[+0.002, +0.031]	no	0.04
1902	TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB+0D	0.724	0.762	+0.038	[+0.025, +0.052]	no	0.17
1903	TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN+0D	0.827	0.828	+0.001	[-0.001, +0.002]	yes	1.00
1903	TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.842	0.842	+0.000	[-0.001, +0.002]	yes	1.00
1904	TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.842	0.843	+0.001	[-0.001, +0.004]	yes	1.00
1905	TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.842	0.842	+0.001	[-0.001, +0.002]	yes	1.00
1905	TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.827	0.828	+0.001	[-0.001, +0.004]	yes	1.00
1906	TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.723	0.755	+0.031	[+0.020, +0.043]	no	0.40
1907	TrQA TinyLL	SINK $\rightarrow$ SINK(+)0D	0.720	0.725	+0.005	[+0.001, +0.009]	yes	1.00
1907	TrQA Mis7B	0D $\rightarrow$ SINK	0.699	0.696	-0.003	[-0.013, +0.007]	yes	0.62
1908	TrQA Mis7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.699	0.693	-0.006	[-0.019, +0.007]	no	0.21
1908	TrQA Mis7B	SINK $\rightarrow$ 0D+SINK	0.696	0.698	+0.002	[-0.007, +0.010]	yes	0.80
1909	TrQA Mis7B	SINK $\rightarrow$ SINK+DEFL	0.696	0.739	+0.043	[+0.026, +0.060]	no	0.00
1909	TrQA Mis7B	0D $\rightarrow$ 0D+1D	0.699	0.692	-0.007	[-0.017, +0.003]	no	0.59
1910	TrQA Mis7B	SINK $\rightarrow$ SINK+1D	0.696	0.686	-0.010	[-0.020, +0.001]	no	0.52
1911	TrQA Mis7B	SINK $\rightarrow$ SINK+ANS	0.696	0.701	+0.005	[-0.010, +0.020]	no	0.00
1911	TrQA Mis7B	PH_0DTOT $\rightarrow$ PH_SINK	0.808	0.812	+0.004	[-0.004, +0.011]	yes	0.88
1912	TrQA Mis7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.812	0.814	+0.002	[+0.000, +0.004]	yes	1.00
1912	TrQA Mis7B	PH_0D $\rightarrow$ PH_ENT	0.775	0.812	+0.037	[+0.023, +0.051]	no	0.09
1913	TrQA Mis7B	0D $\rightarrow$ LOGPROB	0.699	0.719	+0.020	[-0.005, +0.044]	no	0.00
1913	TrQA Mis7B	0D $\rightarrow$ HIDDEN	0.699	0.833	+0.134	[+0.113, +0.155]	no	0.00
1914	TrQA Mis7B	0D $\rightarrow$ LOOKBACK	0.699	0.822	+0.123	[+0.103, +0.143]	no	0.00
1915	TrQA Mis7B	0D $\rightarrow$ LLMCHECK	0.699	0.686	-0.013	[-0.034, +0.008]	no	0.00
1915	TrQA Mis7B	0D $\rightarrow$ LEX	0.699	0.677	-0.023	[-0.047, +0.002]	no	0.00
1916	TrQA Mis7B	0D $\rightarrow$ ROWSTAT	0.699	0.731	+0.032	[+0.017, +0.047]	no	0.03
1916	TrQA Mis7B	0D $\rightarrow$ LEN	0.699	0.635	-0.064	[-0.086, -0.043]	no	0.00
1917	TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.841	0.841	+0.000	[-0.001, +0.001]	yes	1.00
1918	TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.841	0.841	-0.000	[-0.001, +0.001]	yes	1.00
1918	TrQA Mis7B	LEX $\rightarrow$ LEX+0D	0.677	0.723	+0.046	[+0.030, +0.062]	no	0.00
1919	TrQA Mis7B	LEX $\rightarrow$ LEX+SINK	0.677	0.721	+0.045	[+0.029, +0.060]	no	0.00
1919	TrQA Mis7B	LEX $\rightarrow$ LEX+DEFL	0.677	0.744	+0.068	[+0.049, +0.087]	no	0.00
1920	TrQA Mis7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.719	0.752	+0.033	[+0.017, +0.050]	no	0.00
1921	TrQA Mis7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.833	0.834	+0.001	[-0.001, +0.002]	yes	1.00
1921	TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.841	0.843	+0.003	[+0.000, +0.005]	yes	1.00
1922	TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.841	0.846	+0.006	[+0.002, +0.009]	yes	1.00
1922	TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.841	0.843	+0.003	[-0.000, +0.005]	yes	1.00
1923	TrQA Mis7B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.833	0.837	+0.004	[+0.001, +0.006]	yes	1.00
1923	TrQA Mis7B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.718	0.749	+0.031	[+0.016, +0.046]	no	0.00
1924	TrQA Mis7B	SINK $\rightarrow$ SINK(+)0D	0.696	0.700	+0.005	[-0.001, +0.011]	yes	0.98
1925	TrQA Qw7B	0D $\rightarrow$ SINK	0.668	0.714	+0.046	[+0.022, +0.070]	no	0.00
1925	TrQA Qw7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.668	0.672	+0.004	[-0.002, +0.010]	yes	0.98
1926	TrQA Qw7B	SINK $\rightarrow$ 0D+SINK	0.714	0.735	+0.022	[+0.011, +0.033]	no	0.46
1926	TrQA Qw7B	SINK $\rightarrow$ SINK+DEFL	0.714	0.734	+0.020	[+0.008, +0.033]	no	0.27
1927	TrQA Qw7B	0D $\rightarrow$ 0D+1D	0.668	0.668	+0.000	[-0.010, +0.010]	yes	0.59
1927	TrQA Qw7B	SINK $\rightarrow$ SINK+1D	0.714	0.718	+0.004	[-0.004, +0.013]	yes	0.73
1928	TrQA Qw7B	SINK $\rightarrow$ SINK+ANS	0.714	0.781	+0.067	[+0.051, +0.083]	no	0.00
1928	TrQA Qw7B	PH_0DTOT $\rightarrow$ PH_SINK	0.822	0.801	-0.020	[-0.035, -0.006]	no	0.07
1929	TrQA Qw7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.801	0.832	+0.031	[+0.024, +0.039]	no	0.91
1930	TrQA Qw7B	PH_0D $\rightarrow$ PH_ENT	0.798	0.841	+0.044	[+0.032, +0.056]	no	0.33
1930	TrQA Qw7B	0D $\rightarrow$ LOGPROB	0.668	0.847	+0.180	[+0.158, +0.201]	no	0.00
1931	TrQA Qw7B	0D $\rightarrow$ HIDDEN	0.668	0.854	+0.186	[+0.165, +0.207]	no	0.00
1932	TrQA Qw7B	0D $\rightarrow$ LOOKBACK	0.668	0.879	+0.211	[+0.191, +0.231]	no	0.00
1932	TrQA Qw7B	0D $\rightarrow$ LLMCHECK	0.668	0.725	+0.058	[+0.038, +0.078]	no	0.00
1933	TrQA Qw7B	0D $\rightarrow$ LEX	0.668	0.594	-0.074	[-0.101, -0.046]	no	0.00
1933	TrQA Qw7B	0D $\rightarrow$ ROWSTAT	0.668	0.734	+0.067	[+0.053, +0.081]	no	0.10
1934	TrQA Qw7B	0D $\rightarrow$ LEN	0.668	0.595	-0.073	[-0.097, -0.049]	no	0.00
1934	TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.881	0.881	-0.000	[-0.001, +0.000]	yes	1.00
1935	TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.881	0.881	+0.000	[-0.001, +0.001]	yes	1.00
1936	TrQA Qw7B	LEX $\rightarrow$ LEX+0D	0.594	0.678	+0.085	[+0.067, +0.102]	no	0.00
1936	TrQA Qw7B	LEX $\rightarrow$ LEX+SINK	0.594	0.728	+0.134	[+0.113, +0.154]	no	0.00
1937	TrQA Qw7B	LEX $\rightarrow$ LEX+DEFL	0.594	0.678	+0.085	[+0.066, +0.103]	no	0.00
1937	TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.847	0.842	-0.005	[-0.010, +0.001]	yes	1.00
1938	TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.854	0.853	-0.001	[-0.001, -0.000]	yes	1.00
1938	TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
1939	TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
1940	TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.881	0.882	+0.001	[-0.001, +0.003]	yes	1.00
1940	TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.853	0.854	+0.001	[-0.001, +0.002]	yes	1.00
1941	TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.847	0.846	-0.000	[-0.003, +0.002]	yes	1.00
1942	TrQA Qw7B	SINK $\rightarrow$ SINK(+)0D	0.713	0.732	+0.019	[+0.009, +0.029]	no	0.60
1943								

Table 13: **Key paired comparisons** (ΔAUC). “X–Y” is $\text{AUC}(\text{X}) - \text{AUC}(\text{Y})$; “X|Y” is $\text{AUC}(\text{Y}+\text{X}) - \text{AUC}(\text{Y})$ under early fusion; “X $\oplus$ Y” is the same under late fusion (stacked out-of-fold logits). $P_{0,h}$: per-head MST total. Defl.: sink-deflated 0D+1D PH; NT: all non-topological features. $\equiv$: equivalent within ± 0.015 (90% interval inside the margin); *: 95% interval excludes 0. Intervals and TOST power for every cell are in Appendix G.

Setting	Sink–0D	0D Sink	$P_{0,h}$ Sink _h	Defl. Sink	1D 0D	0D NT	0D $\oplus$ NT	Defl. NT
TQA Qw3B	+0.001 $\equiv$	+0.003 $\equiv$	+0.009 $\equiv$	+0.029*	+0.033*	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw1.5B	–0.003 $\equiv$	+0.012*	+0.011 $\equiv$	+0.034*	–0.008 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Phi-3	+0.005 $\equiv$	+0.006 $\equiv$	0.000 $\equiv$	+0.033*	–0.003 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA TinyLL	–0.010	+0.020*	+0.003 $\equiv$	+0.045*	+0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA SmoLLM	+0.028*	+0.030*	+0.106*	+0.041*	+0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA Mis7B	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.055*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Mis7B (chat)	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.067*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Qw7B	+0.070*	+0.027*	+0.060*	+0.038*	+0.021*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw3B	+0.018*	+0.028*	+0.004 $\equiv$	+0.058*	+0.015*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	+0.031*	+0.012*	+0.008 $\equiv$	+0.052*	+0.026*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	–0.015*	+0.021*	+0.004 $\equiv$	+0.002	–0.003 $\equiv$	+0.001 $\equiv$	+0.002 $\equiv$	0.000 $\equiv$
TrQA Qw1.5B	–0.009	+0.010	+0.004 $\equiv$	+0.012	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TrQA Phi-3	–0.004 $\equiv$	+0.006 $\equiv$	+0.002 $\equiv$	+0.029*	–0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.002 $\equiv$
TrQA TinyLL	+0.001 $\equiv$	+0.004 $\equiv$	–0.001 $\equiv$	+0.014	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Mis7B	–0.003 $\equiv$	+0.002 $\equiv$	+0.002 $\equiv$	+0.043*	–0.007	0.000 $\equiv$	+0.003 $\equiv$	0.000 $\equiv$
TrQA Qw7B	+0.046*	+0.022*	+0.031*	+0.020*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$

H INTERVENING ON THE SINK

Proposition 1 predicts that TOHA should track the response sink score more closely as the sink strengthens. We test this by intervening on the sink directly. For Qwen2.5-1.5B and TinyLlama on TruthfulQA, we add a constant $b \in \{-2, 0, 2, 4\}$ to the first token’s attention logit in every head and recompute all per-head quantities (Table 16). The intervention leaves the identity intact (no violations in any run). It moves the structure the proposition says matters. Suppressing the sink ($b=-2$) lowers the share of prompt-coned head graphs from 58% to 37% (Qwen) and from 74% to 63% (TinyLlama). It lowers the share where the sink is every response token’s top prompt token from 45% to 21% and from 63% to 21%, and the median rank correlation between d and the response sink score from 0.893 to 0.605 and from 0.967 to 0.406. Strengthening it ($b=4$) raises the correlation to 0.962 and 0.996. In TinyLlama all three quantities increase monotonically with b . In Qwen they peak at $b=2$ and fall slightly at $b=4$. TOHA’s own AUC barely moves (0.65 to 0.67 and 0.65 to 0.70). Its gap to the sink-score version is not monotone in b , so the intervention supports the mechanism at the level of the attention graph, not a dose-response in detection accuracy.

Table 16: **Sink-bias intervention** (teacher-forced TruthfulQA; labels unchanged). A constant b is added to the attention logit of key 0 for every query after token 0, in every head and layer. $\delta_P=0$: share of prompt-coned head graphs. Sink top: share of head graphs in which token 0 is every response token’s most-attended prompt token. ρ : median per-head rank correlation between TOHA’s d and the response sink score. Remaining columns: AUCs as in Table 3.

Model	b	$\delta_P=0$	sink top	$\rho(d, 1 - \bar{a}_0)$	TOHA	TOHA[$\bar{a}_0$]	TOHA _{sup}	$\bar{a}_{0,\text{sup}}$
Qwen2.5-1.5B	–2	37%	21%	0.605	0.667	0.624	0.843	0.849
Qwen2.5-1.5B	0	58%	45%	0.893	0.646	0.628	0.858	0.857
Qwen2.5-1.5B	+2	72%	60%	0.970	0.667	0.637	0.857	0.857
Qwen2.5-1.5B	+4	69%	56%	0.962	0.647	0.640	0.839	0.838
TinyLlama-1.1B	–2	63%	21%	0.406	0.698	0.626	0.845	0.834
TinyLlama-1.1B	0	74%	63%	0.967	0.666	0.618	0.845	0.848
TinyLlama-1.1B	+2	81%	75%	0.992	0.652	0.610	0.848	0.848
TinyLlama-1.1B	+4	86%	83%	0.996	0.665	0.616	0.844	0.842

I MISTRAL-7B

Mistral-7B-Instruct-v0.2 is the most sink-dominated model in this study, and the reduction is correspondingly tightest there. Under its `[INST]` format it places 0.75 of its attention on token 0 — more than any smaller model — and 98.2% of its 52,288 layer graphs are *exactly* coned ($\delta_0 = 0$), with 0 violations of any bound in Theorem 1.

An earlier accuracy-level audit of this model reported that 99.99% of its 1D features are exactly zero and that the 0D and 0D+1D banks are equivalent, and inferred from Theorem 1(c) that near-universal coning would account for it — without checking δ_0 directly. Both halves now hold on the graphs themselves: 1D persistence is exactly zero on 99.99% of them, and $\delta_0 = 0$ is verified on 98.2%. The length-normalized rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ has median 1.000 over layers and never falls below 0.994, so on this model 0D total persistence is, up to rank, an affine function of sink attention in every layer. Per head, 81% of its graphs are exactly prompt-coned, the median $\rho(d, \bar{\pi})$ is 0.998, and Proposition 1 has 0 violations.

J RELATION TO ACCURACY-LEVEL AUDITS, AND COST

Audits of attention-graph PH that stop at classification accuracy observe four facts that the theory now explains: (i) 0D persistence equals an MST statistic (Fact 1); (ii) 1D persistence adds little beyond 0D (Table 13, column “1D | 0D”), which Theorem 1 now explains; (iii) an earlier audit reported that the 1D bank of Mistral-7B under its `[INST]` format is 99.99% zeros, which is what exact coning would produce (not yet re-verified with this pipeline; Appendix I); (iv) HaluEval answers are separable from text alone.

Cost. Median per-matrix wall-clock time on real attention from TinyLlama-1.1B-Chat-v1.0 (sink features / δ_0 / dense MST / ripser 0D / ripser 0D+1D) is 0.001/0.02/0.10/0.02/0.02 ms at $N=32$; 0.001/0.03/0.21/0.03/0.03 ms at $N=64$; 0.001/0.07/0.42/0.07/0.06 ms at $N=128$; 0.001/0.25/0.90/0.18/0.18 ms at $N=256$. On i.i.d. random matrices of the same size, ripser 0D+1D takes 0.171 ms ($N=32$), 3.46 ms ($N=64$), 115 ms ($N=128$), 4,631 ms ($N=256$). On real attention the same computation is four orders of magnitude cheaper at $N=256$, consistent with Theorem 1: sink-dominated graphs have few, short 1D bars to compute. Sink features, read off one column, are one to two orders of magnitude cheaper still.